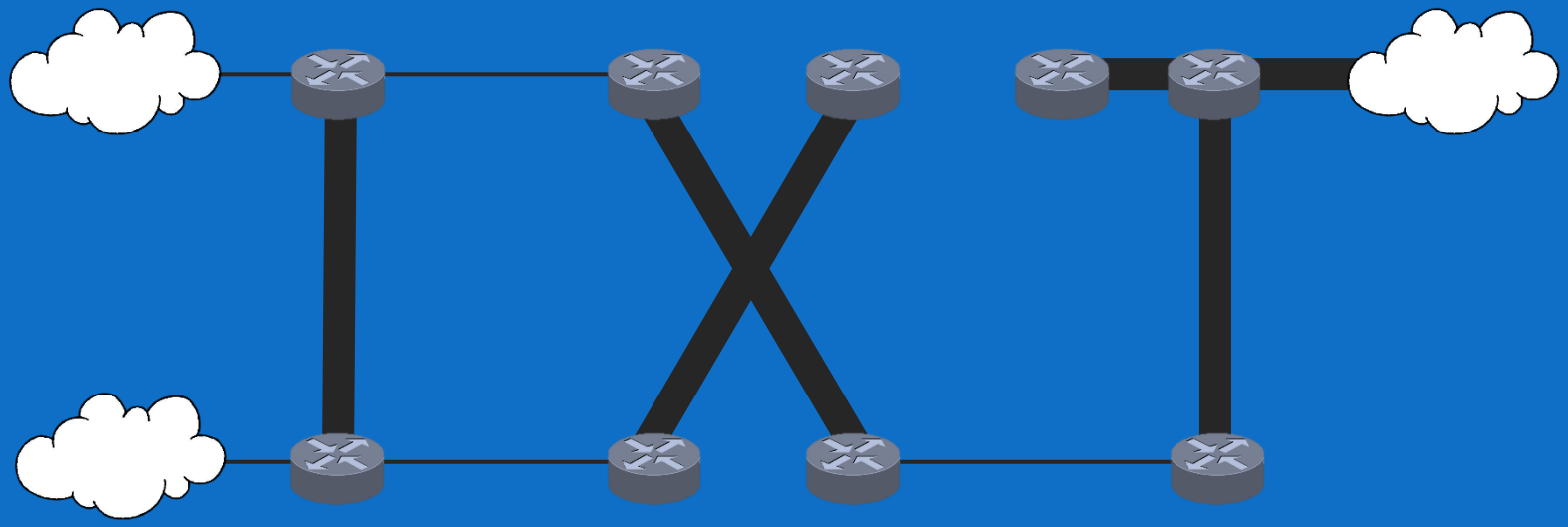
Marcos Postigo-Boix
You can also consult the slides in the "Metacurs",
but this is the version we will follow in the laboratory

Introduction to Networks

(a.k.a. IXT)



Lab 3

Ethernet
Local Area Networks (LAN)



Workstation Cabling



Objectives

You will be able to…

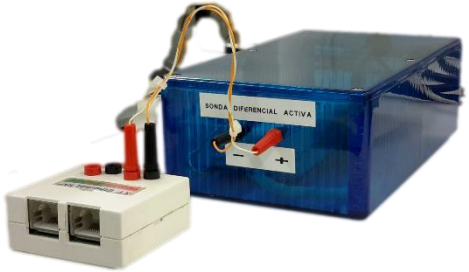
…identify the Ethernet switch



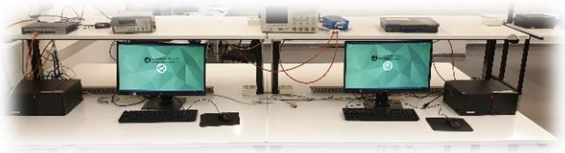


…connect two PCs using an Ethernet switch

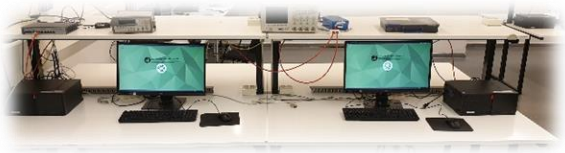
…identify the differential probe

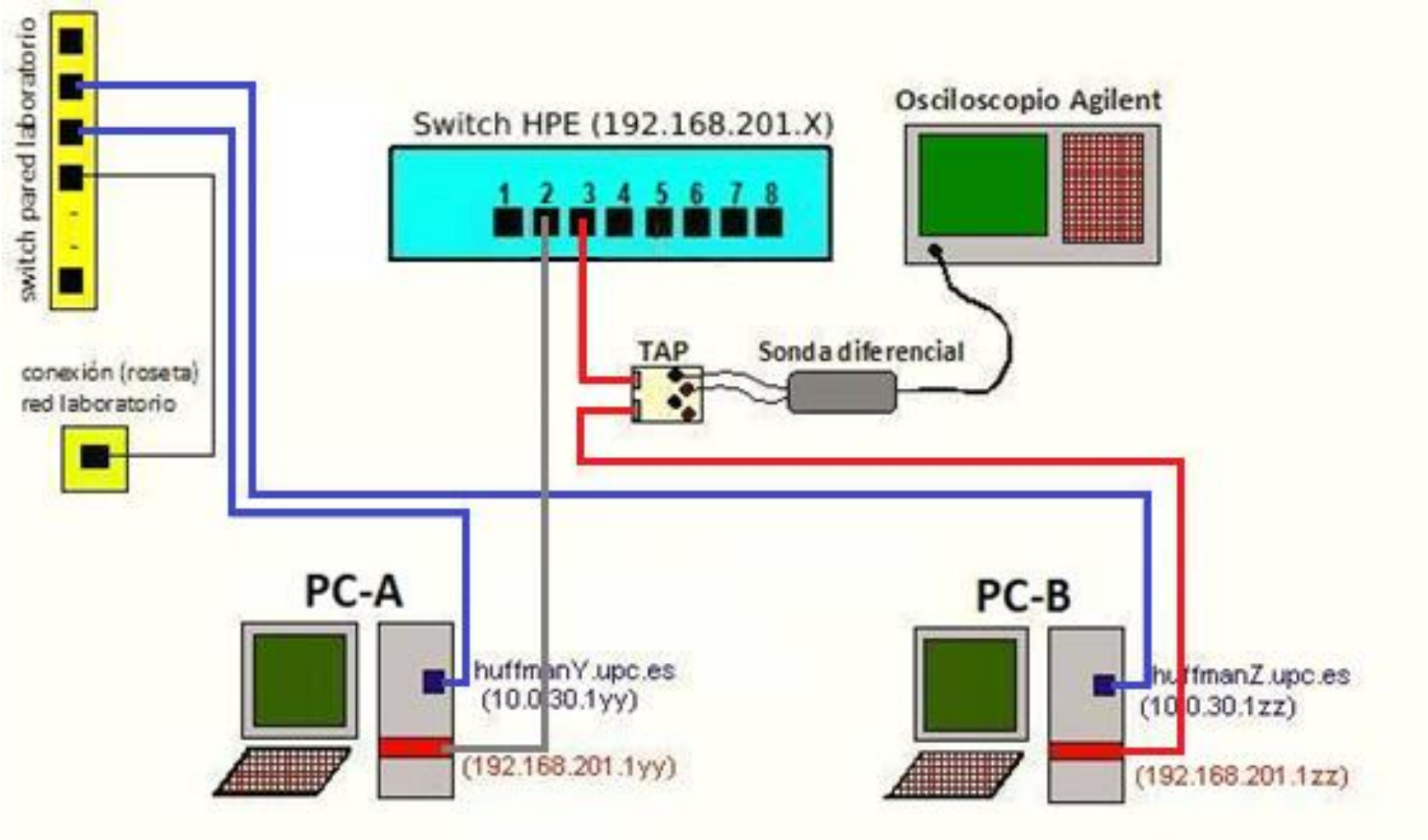


Workstation

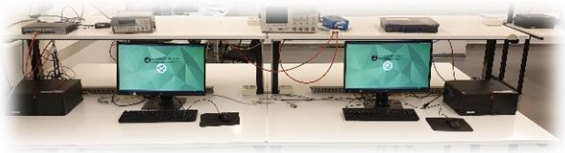


Workstation





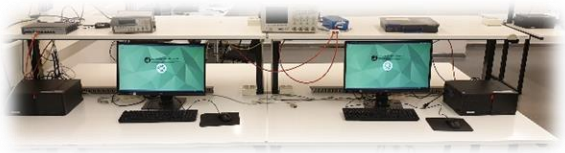
Workstation



Differential Probe

Tap

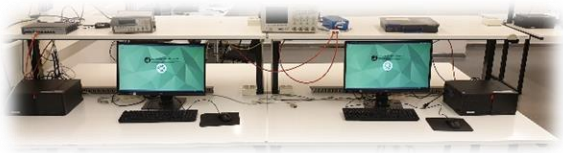
Workstation



Switch
HPE

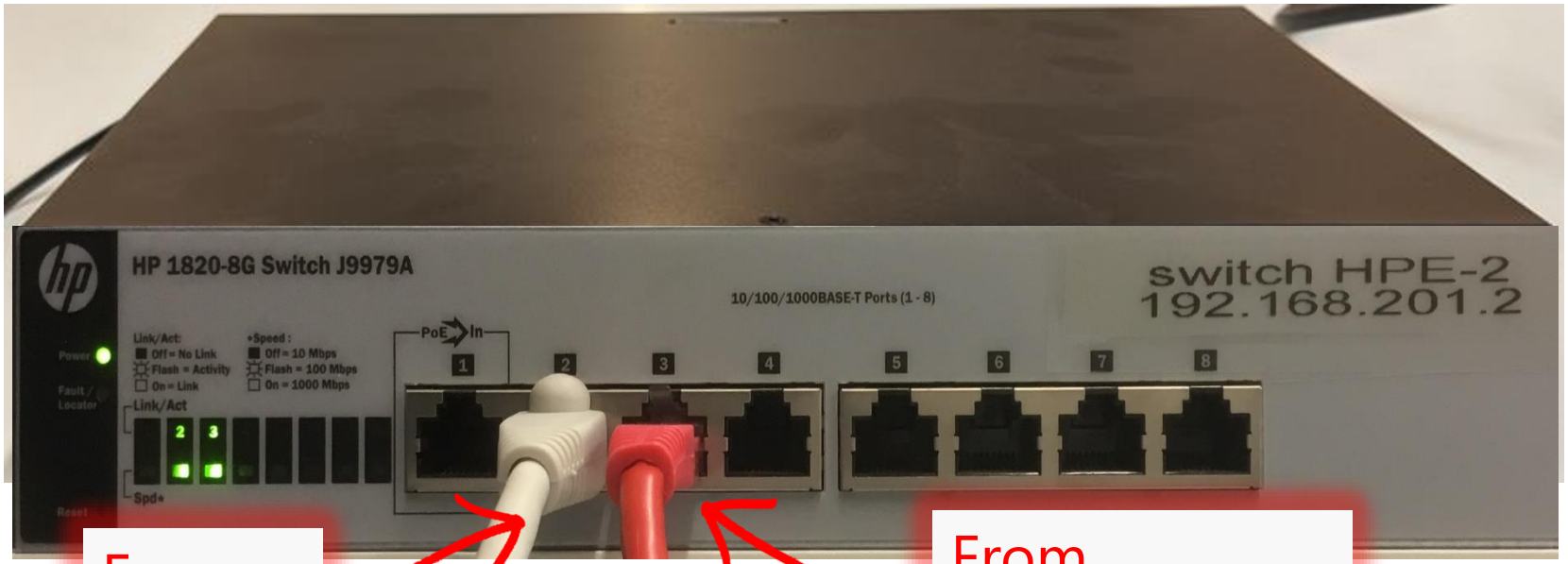


Workstation



Make sure
cables are connected!

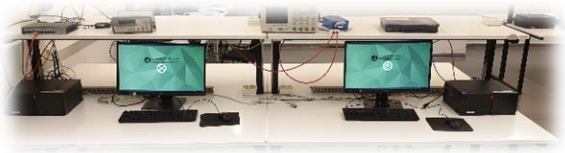
Switch
HPE



From
PC-A

From
TAP/PC-B

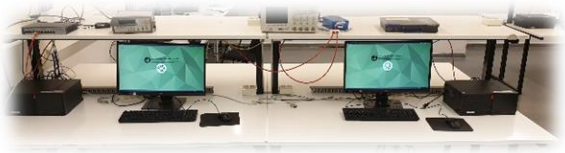
Workstation



Switch
HPE

Plug in

Workstation

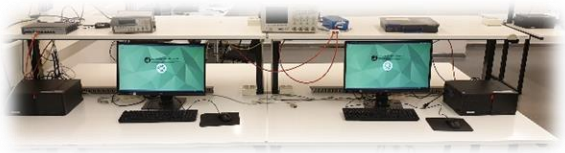


Connect cables to the TAP

From Switch

From PC-B

Workstation



Connect probe
to the Oscilloscope

Channel 1

From
Differential Probe

Lab Exercises

Experiments 1 to 3



Lab Exercises
Experiment 1

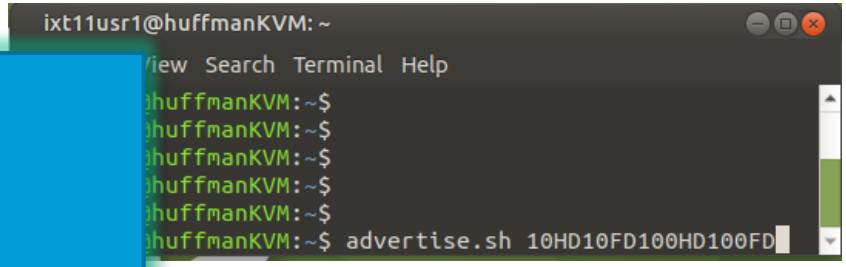
Auto-negotiation Pulses Analysis

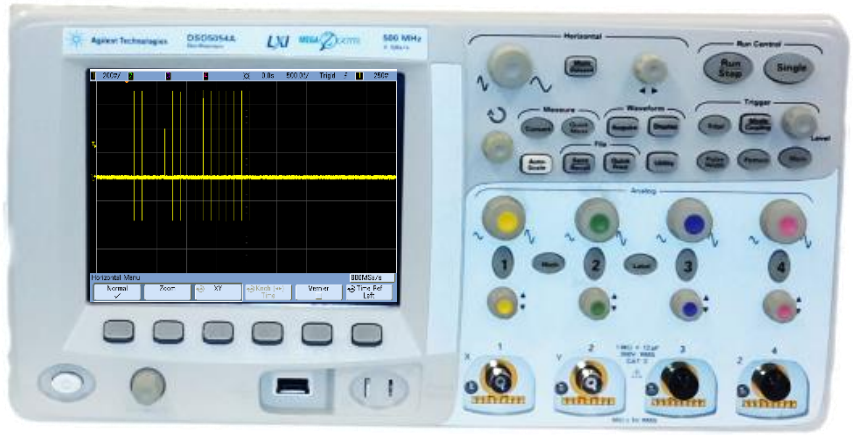


Objectives

You will be able to…

…configure
different link capabilities
advertised by an Ethernet interface

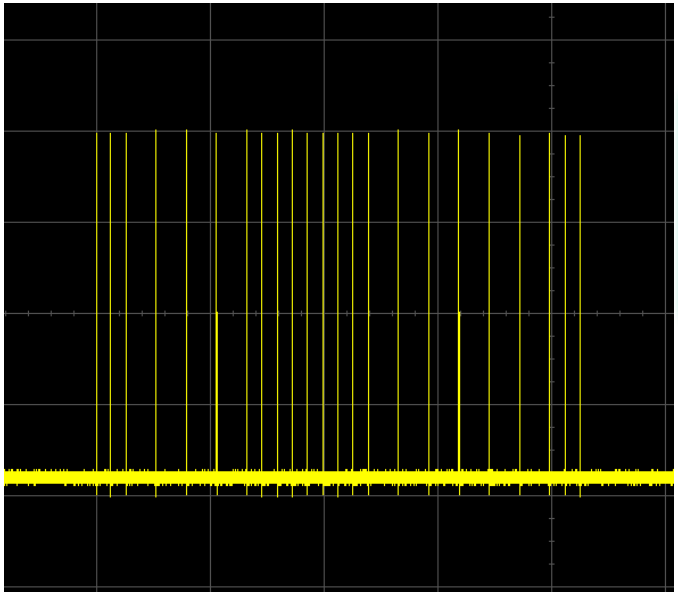




…configure
the oscilloscope to
capture FLP bursts

Objectives

You will be able to…



…examine
an FLP to identify the link codeword

…interpret
the information included in a link codeword

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
S0	S1	S2	S3	S4	A0	A1	A2	A3	A4	A5	A6	XNP	RF	Ack	NP

Experiment 1

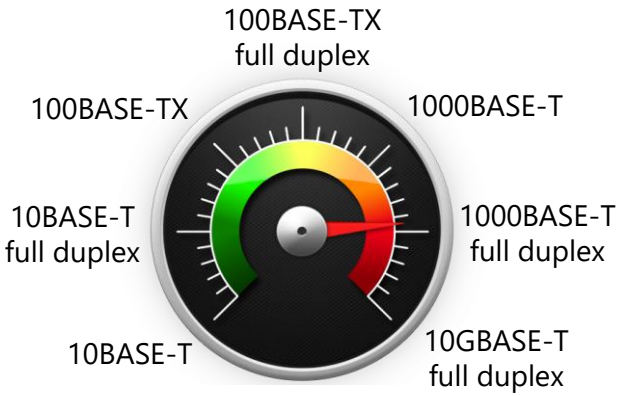
Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation



Information exchange between connected network adapters



...to operate on the best shared mode

Experiment 1

Auto-negotiation Pulses Analysis

Ethernet

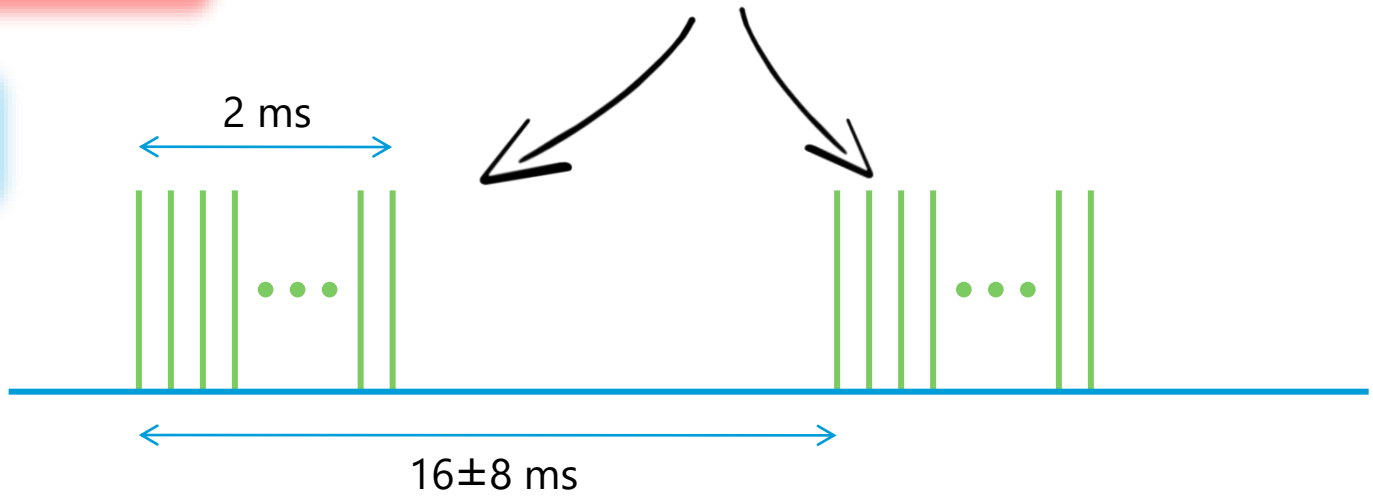
Auto-negotiation



Fast Link Pulse

They transmit data

FLP Bursts



Experiment 1

Auto-negotiation Pulses Analysis

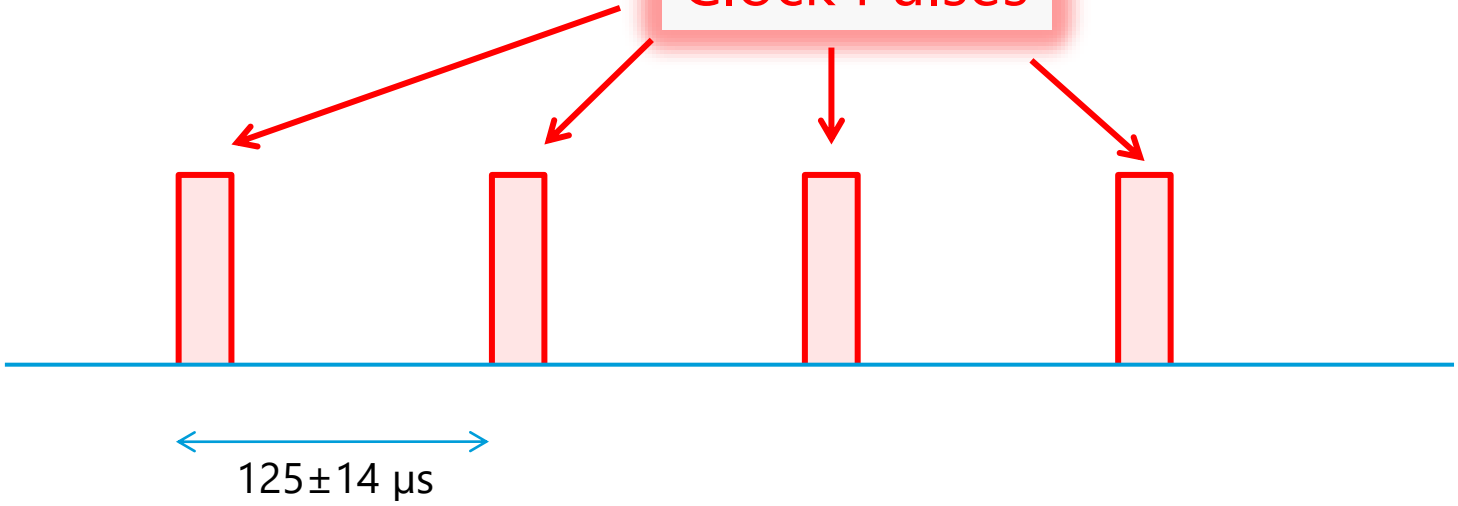
Ethernet

Auto-negotiation

FLP Bursts



Clock Pulses



Experiment 1

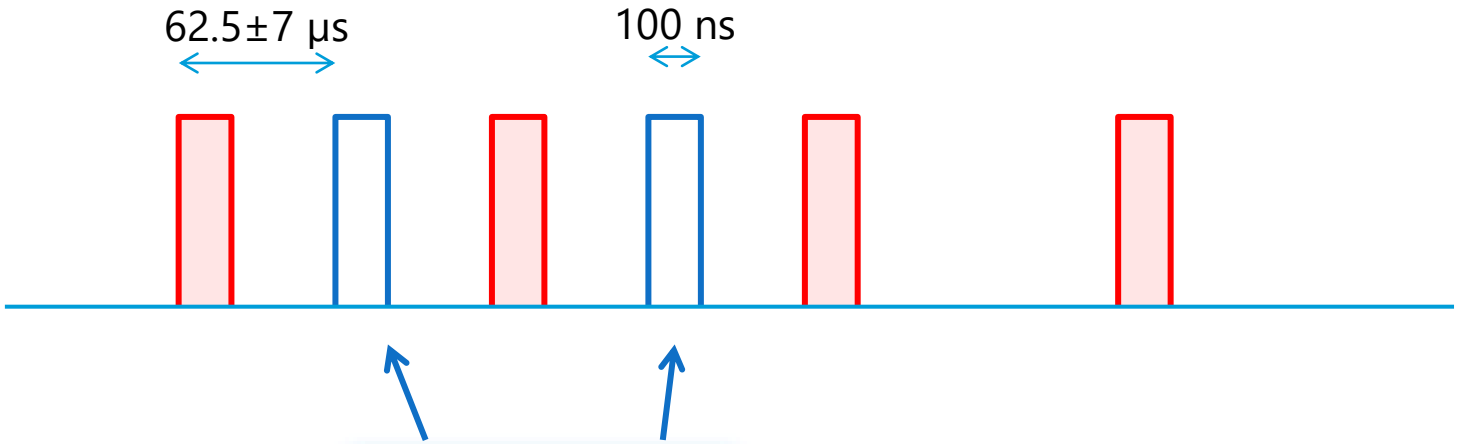
Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation

FLP Bursts





1 pulse present

0 pulse absent

Experiment 1

Auto-negotiation Pulses Analysis

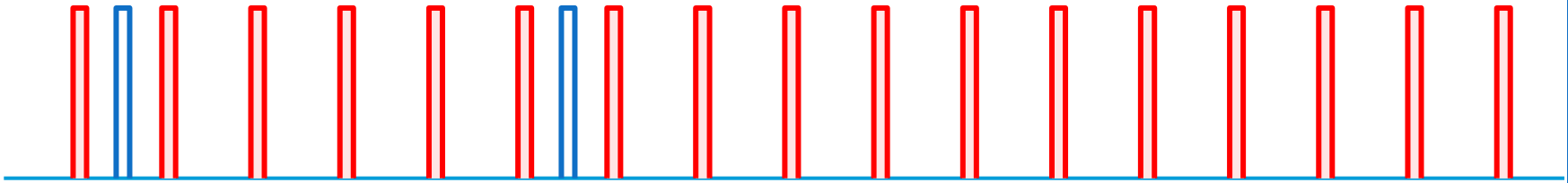
Ethernet

Auto-negotiation

FLP Bursts



Pulse	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
Encoding		D0		D1		D2		D3		D4		D5		D6		D7		D8		D9		D10		D11		D12		D3		D14		D15	
Data		**1**		0		0		0		0		**1**		0		0		0		0		0		0		0		0		0		0	



Experiment 1

Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation



Data to transmit

Base link codeword

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
S0	S1	S2	S3	S4	A0	A1	A2	A3	A4	A5	A6	R	RF	Ack	NP

Experiment 1

Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation



Base link codeword

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
S0	S1	S2	S3	S4	A0	A1	A2	A3	A4	A5	A6	XNP	RF	Ack	NP

Selector Field

Standard in use

D0	D1	D2	D3	D4
1	0	0	0	0

IEEE 802.3 Ethernet

Experiment 1

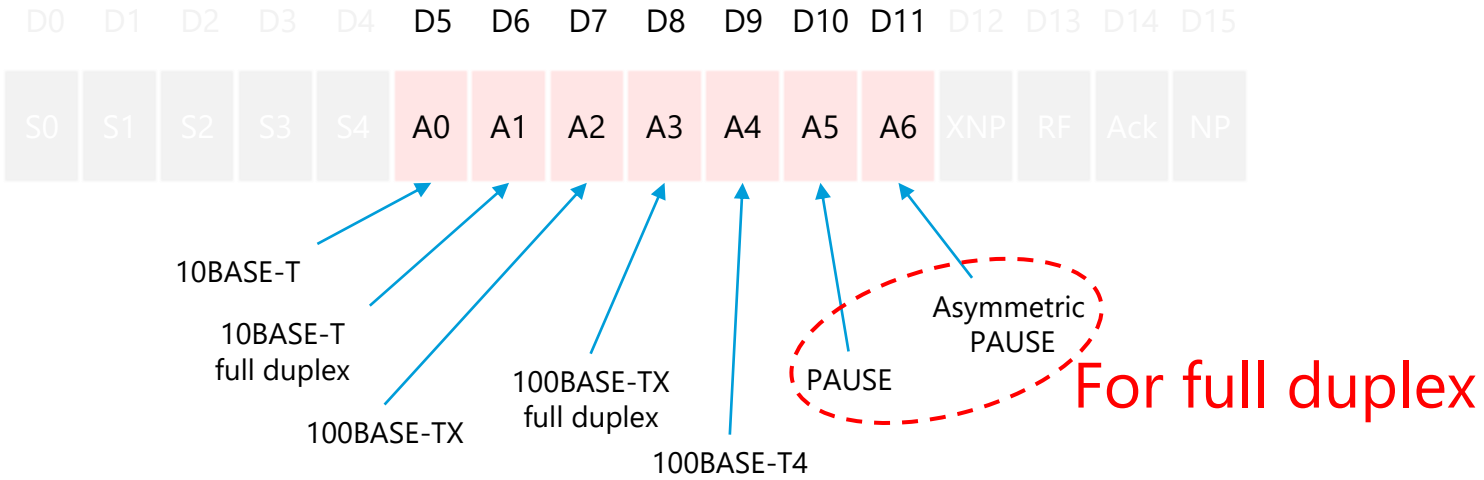
Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation



Base link codeword



Technology Ability Field

Supported Technologies

Experiment 1

Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation



Base link codeword

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
S0	S1	S2	S3	S4	A0	A1	A2	A3	A4	A5	A6	R	RF	Ack	NP

Reserved

For future uses

Extended Next Pages (XNP) is proposed to support transmissions over 10G

Experiment 1

Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation



Base link codeword

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
S0	S1	S2	S3	S4	A0	A1	A2	A3	A4	A5	A6	XNP	RF	Ack	NP

Remote Fault

Fault condition has occurred

Experiment 1

Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation



Base link codeword

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
S0	S1	S2	S3	S4	A0	A1	A2	A3	A4	A5	A6	XNP	RF	Ack	NP

Acknowledge

Successfully received
its Link Partner’s link codeword

Set to 1 after receiving at least
3 consecutive and identic FLP Bursts

Transmitted
6-8 times

Experiment 1

Auto-negotiation Pulses Analysis

Ethernet

Auto-negotiation



Base link codeword

D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
S0	S1	S2	S3	S4	A0	A1	A2	A3	A4	A5	A6	XNP	RF	Ack	NP

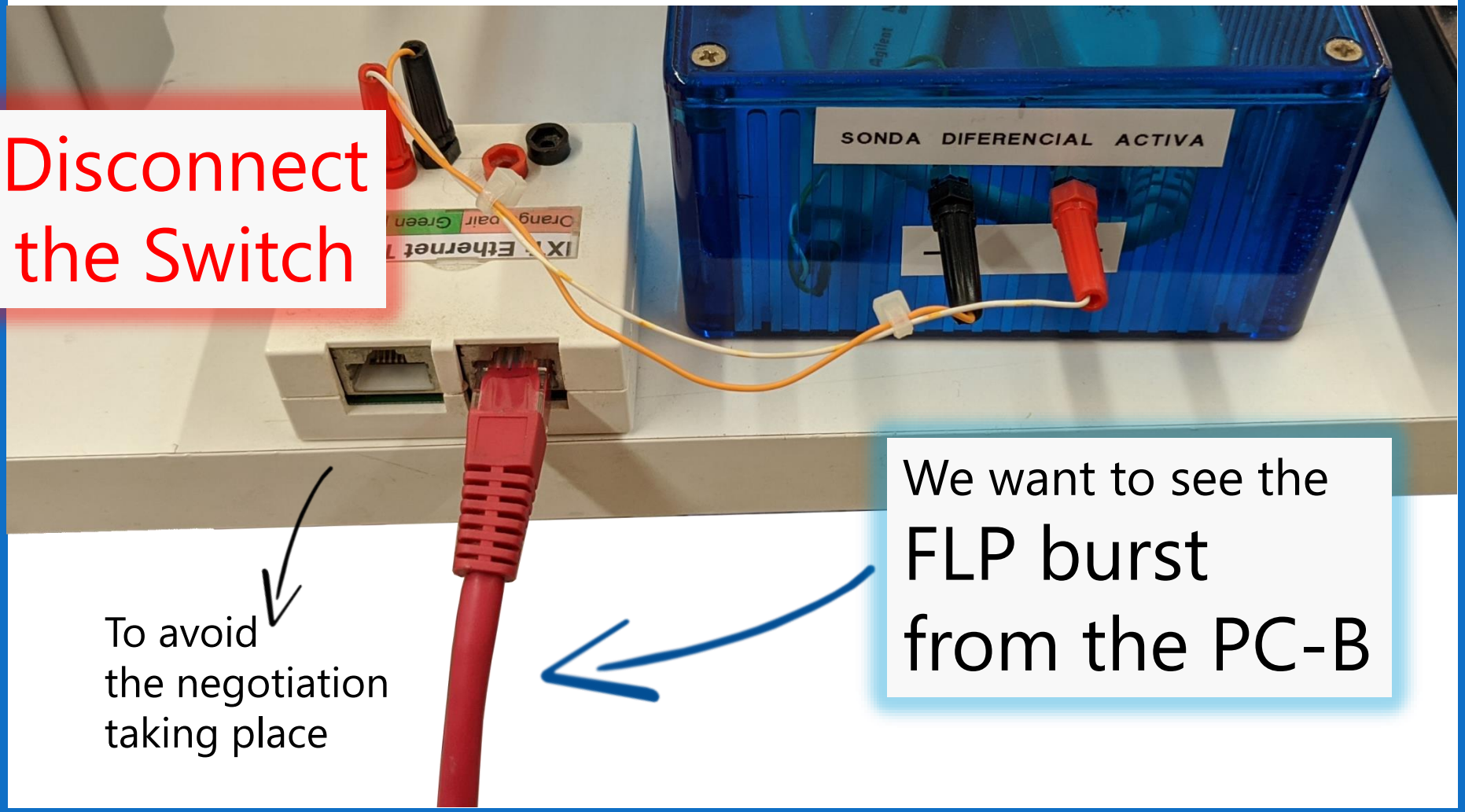
Next Page

Indicates the intention of sending additional link codewords

Only if NP=1 in both devices

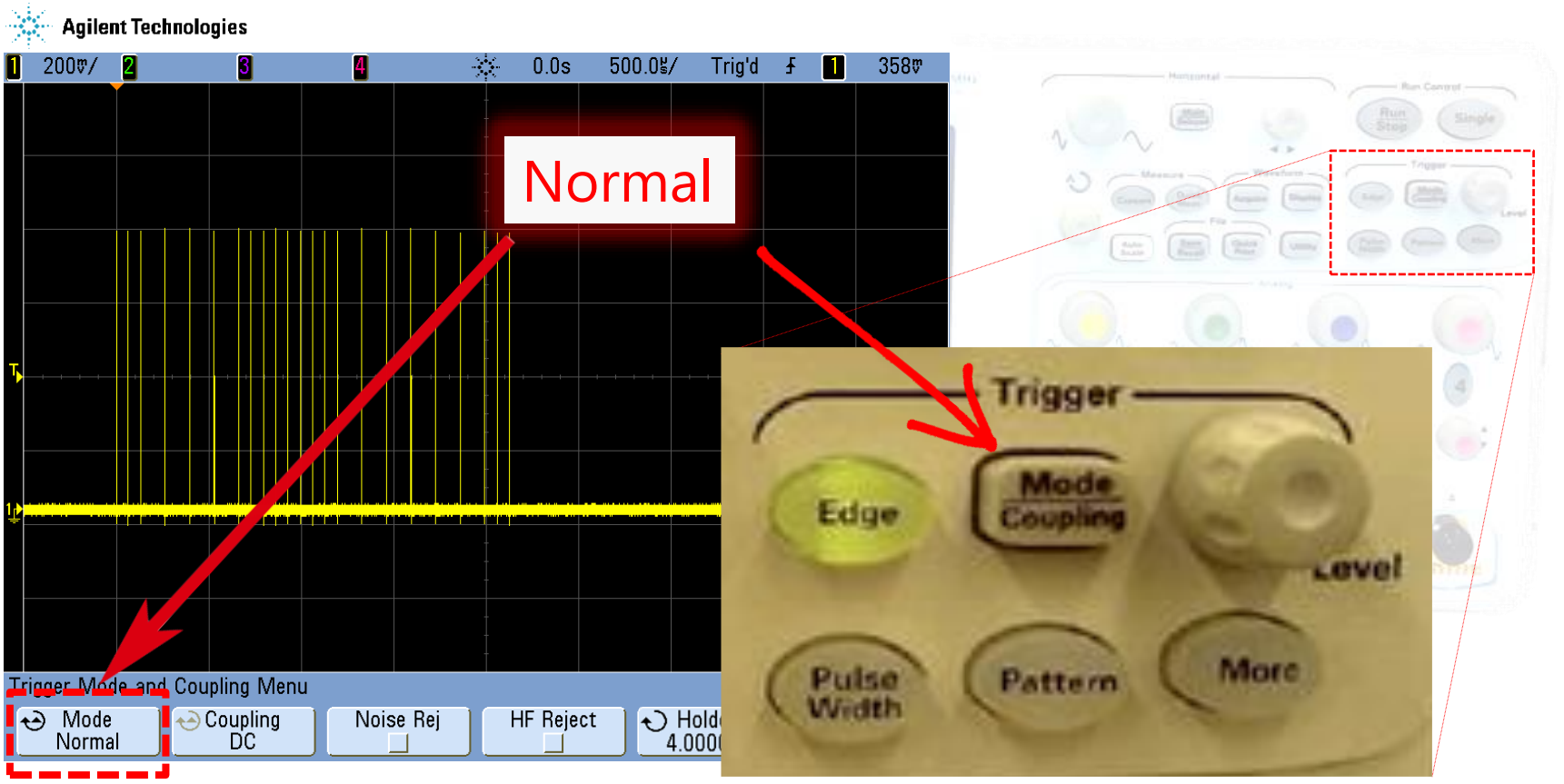
Experiment 1

Auto-negotiation Pulses Analysis



Experiment 1

Auto-negotiation Pulses Analysis



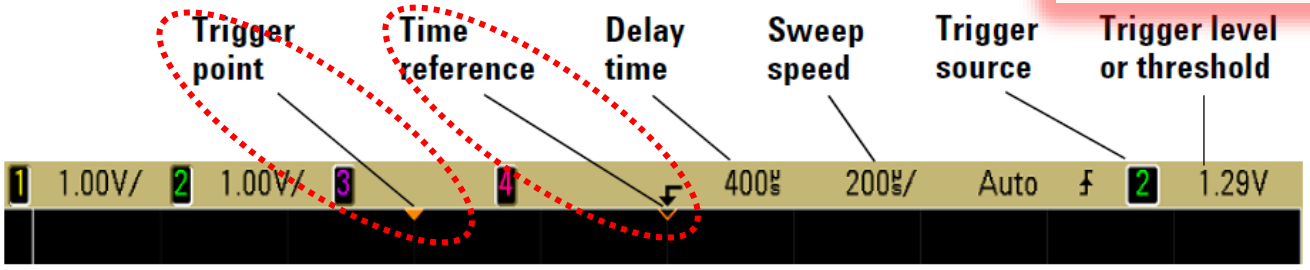
Experiment 1

Auto-negotiation Pulses Analysis



Make the trigger point indicator to overlay the time reference indicator

Delay time set to 0

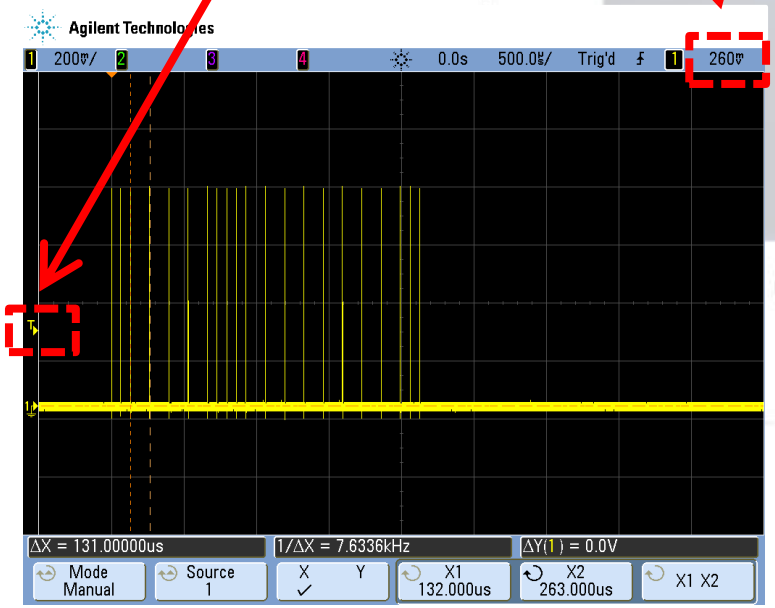


Experiment 1

Auto-negotiation Pulses Analysis

Trigger level
~260 mV

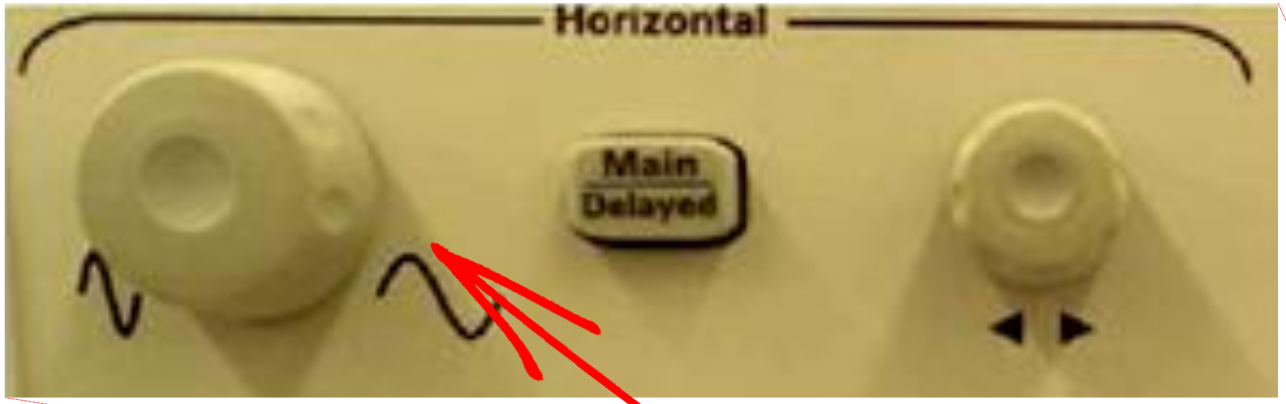
Below signal level

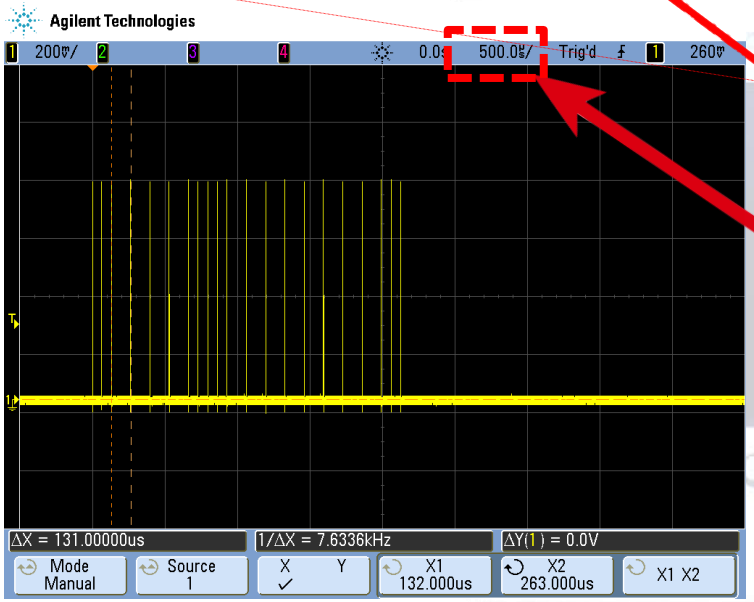




Experiment 1

Auto-negotiation Pulses Analysis



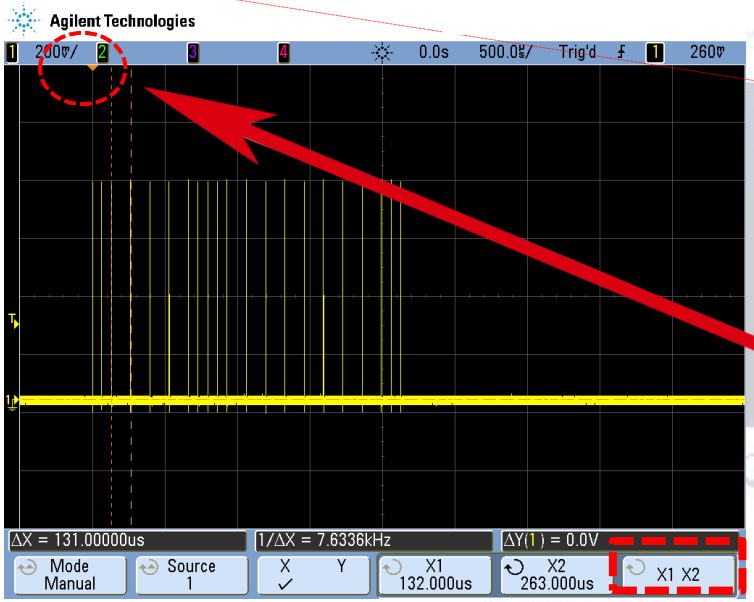


500 µs

Experiment 1

Auto-negotiation Pulses Analysis





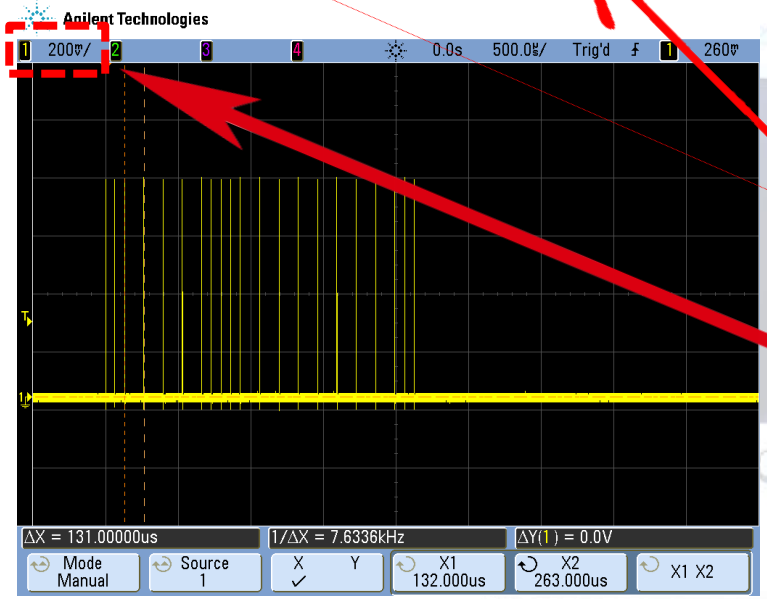
To see a whole FLP burst

Left
Time Reference

Experiment 1

Auto-negotiation Pulses Analysis





200 mV/div

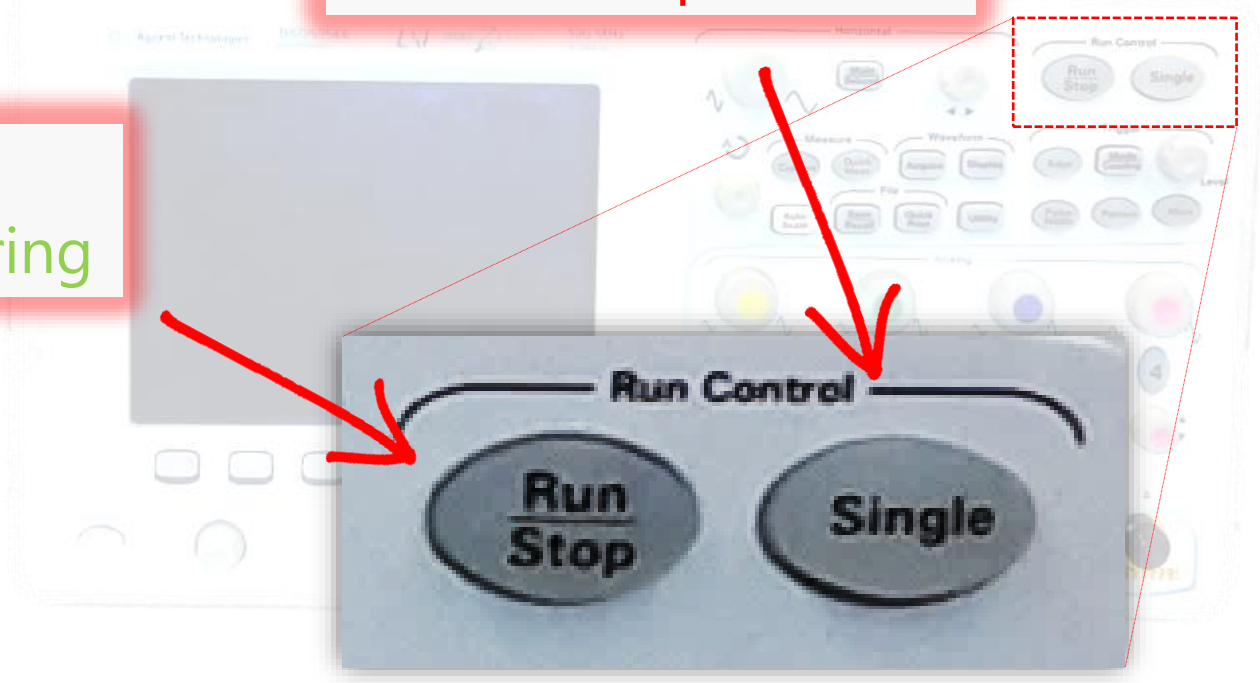
To see a whole FLP burst

Experiment 1

Auto-negotiation Pulses Analysis

Use single acquisition to fix the capture

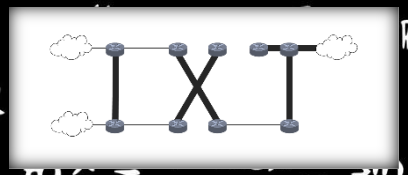
RED stopped
GREEN capturing



Exercises

Experiment 1

Auto-negotiation Pulses Analysis



Capture an FPL Burst

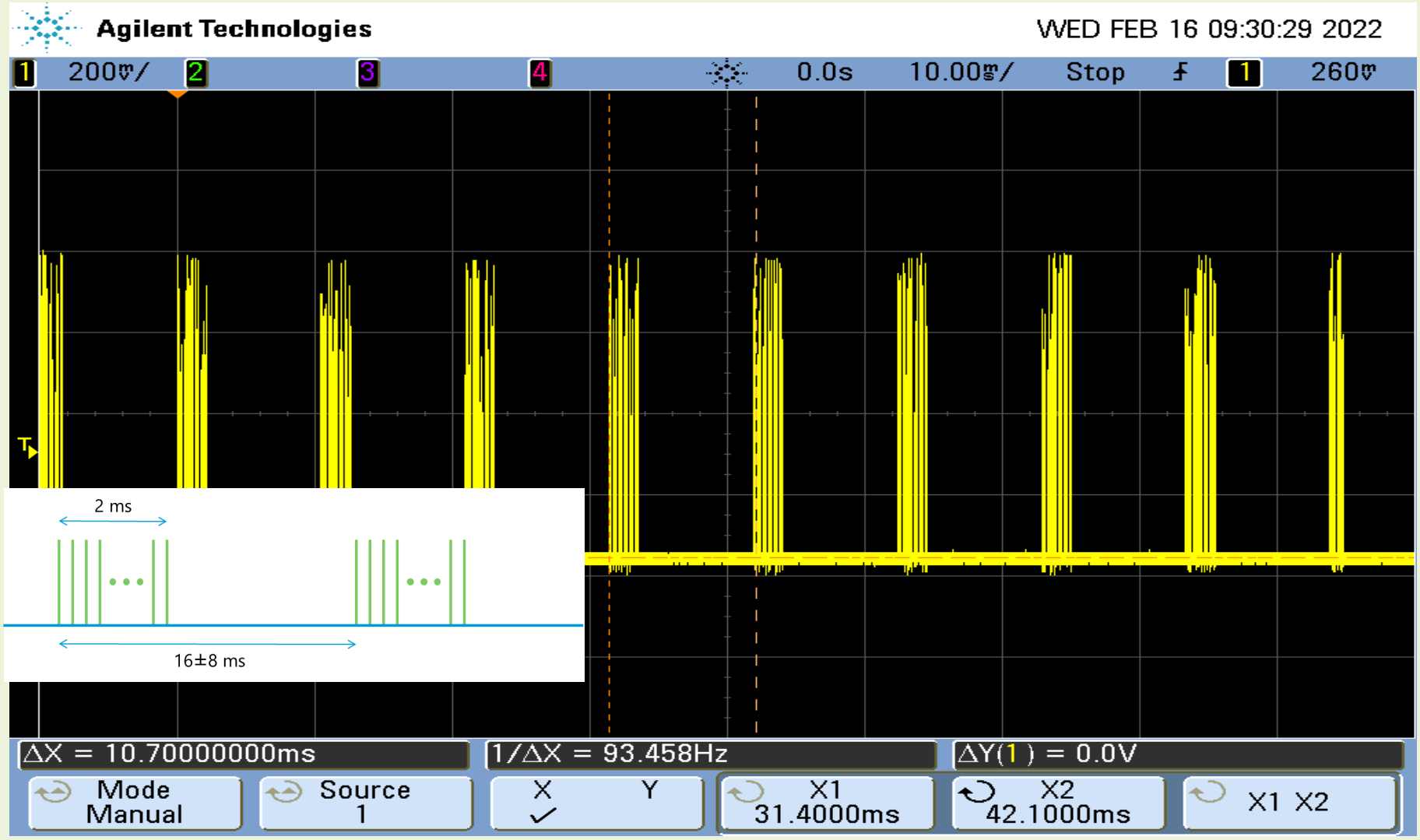
Identify the Link Codeword

Identify the information included in the Codeword

Obtain a copy of the capture from a web browser
`http://@IP`
@IP on top of the oscilloscope

Experiment 1
Auto-negotiation Pulses Analysis

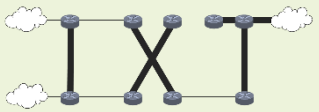
Capture an FPL Burst

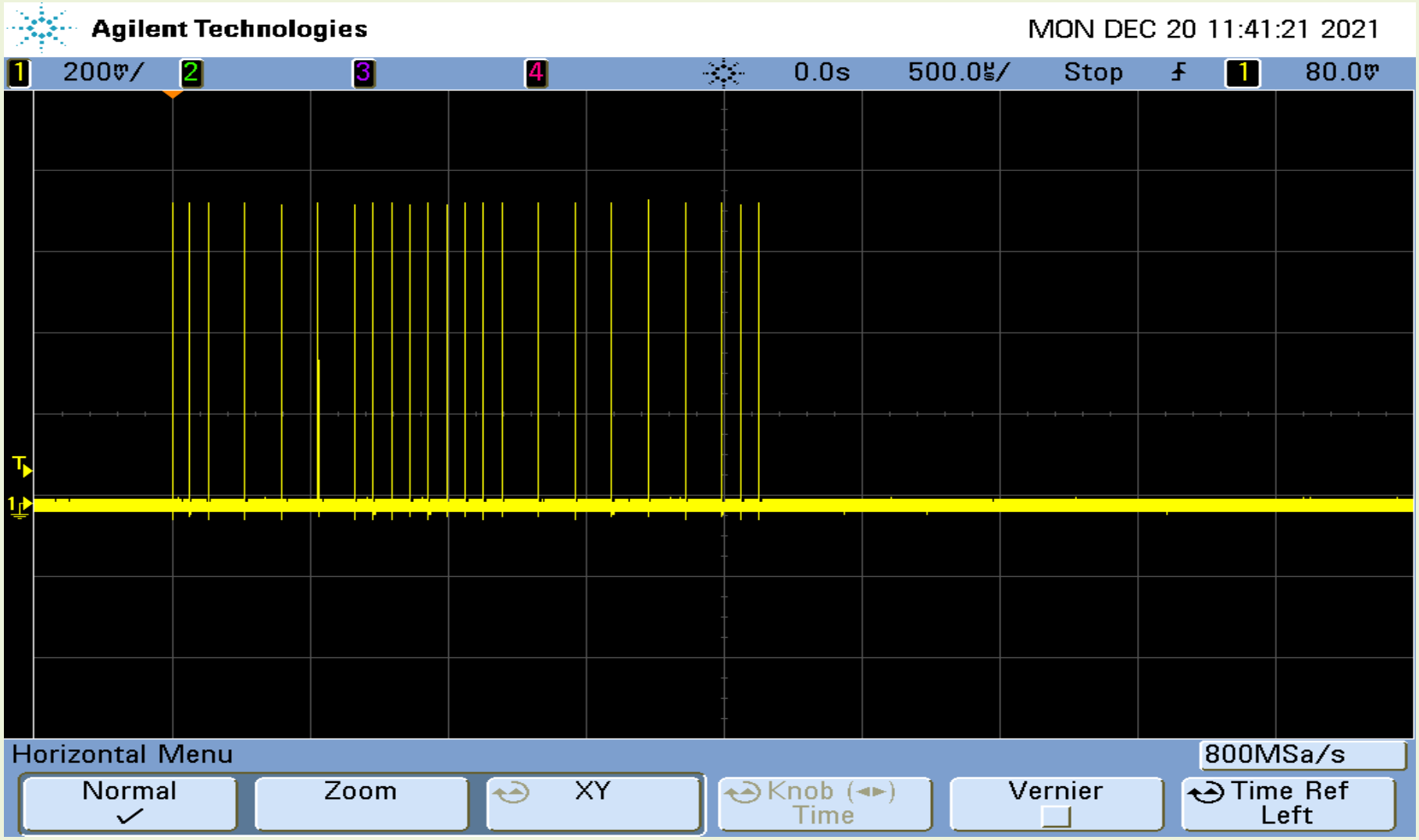


Experiment 1

Auto-negotiation Pulses Analysis

Capture an FPL Burst

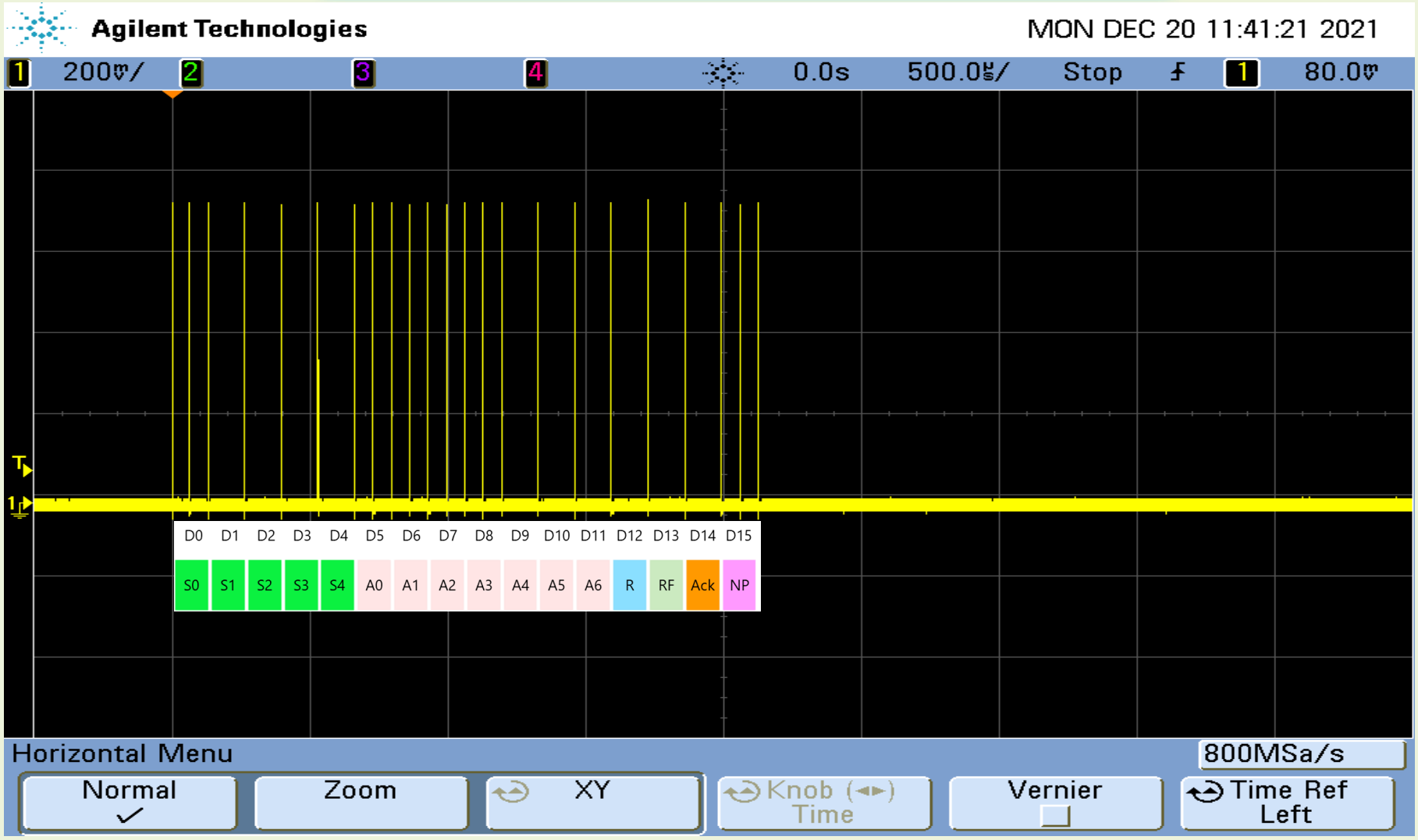




Experiment 1

Auto-negotiation Pulses Analysis

Identify the Link Codeword

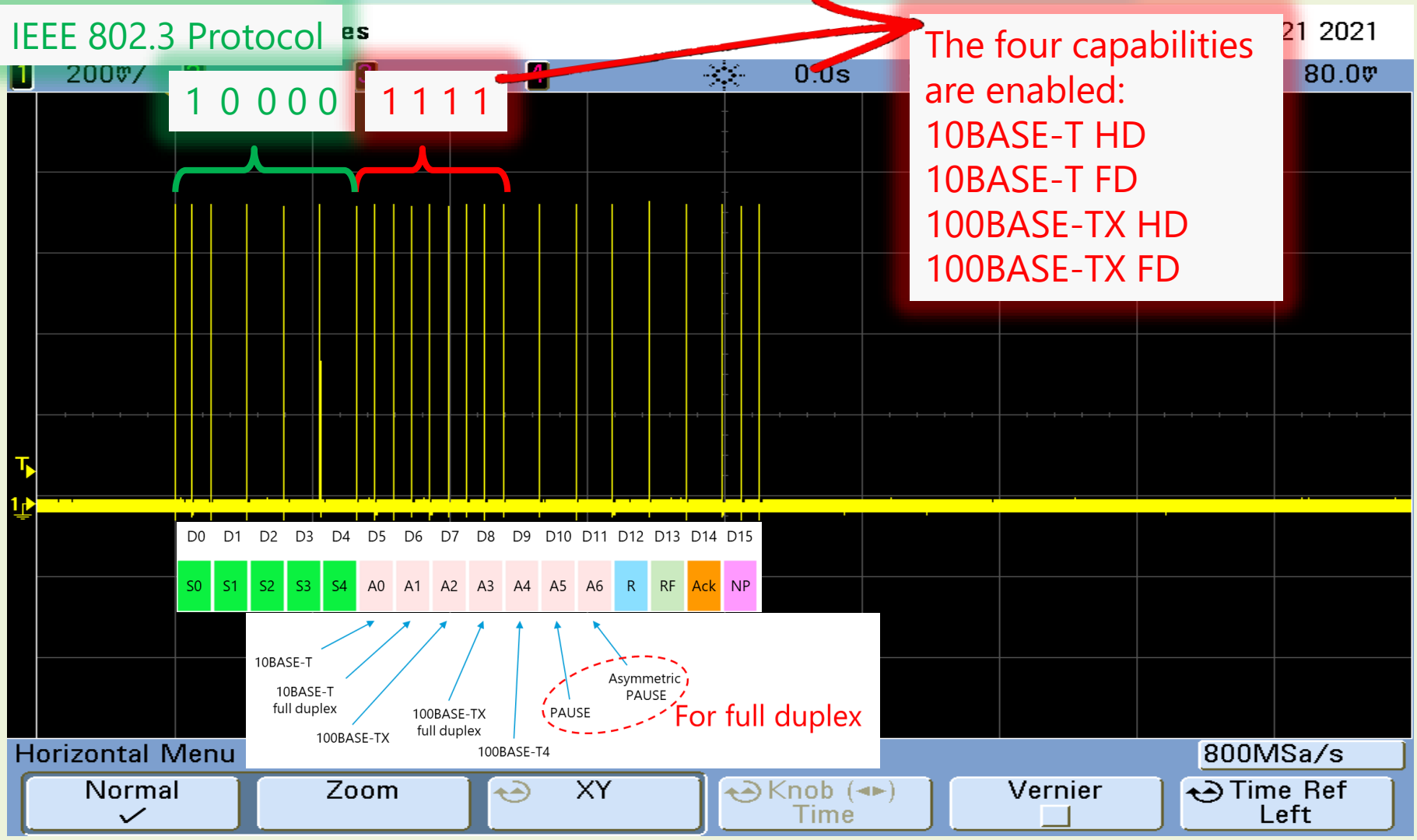


Experiment 1

Auto-negotiation Pulses Analysis

Identify the information included in the Codeword

IEEE 802.3 Protocol



Experiment 1
Auto-negotiation Pulses Analysis

Change the
Advertised Link Modes...

Enabling
Disabling

10BASE-T HD
10BASE-T FD
100BASE-TX HD
100BASE-TX FD

...executing the script `advertise.sh`

Observe those changes in the Link Codeword

Obtain a copy of the capture from a web browser
`http://@IP`
@IP on top of the oscilloscope

Setting of Advertised Link Modes

```
File  Edit  View  Search  Terminal  Help
iotprofe1@huffman4:~$ ethtool eth1_real
Settings for eth1_real:
        Supported ports: [ TP MII ]
        Supported link modes:   10baseT/Half 10baseT/Full
                                100baseT/Half 100baseT/Full
                                1000baseT/Half 1000baseT/Fu
        Supported pause frame use: Symmetric Receive-only
        Supports auto-negotiation: Yes
        Supported FEC modes: Not reported
        Advertised link modes:  10baseT/Half 10baseT/Full
        Advertised pause frame use: No
        Advertised auto-negotiation: Yes
        Advertised FEC modes: Not reported
        Speed: Unknown!
        Duplex: Unknown! (255)
        Port: MII
        PHYAD: 0
        Transceiver: internal
        Auto-negotiation: on
```

We want to see the **Current port settings**

Setting of Advertised Link Modes

```
ixt11usr1@huffmanKVM: ~
File  Edit  View  Search  Terminal  Help
ixt11usr1@huffmanKVM:~$ advertise.sh
Paràmetre invàlid

/usr/local/bin/advertise.sh PAR
    PAR:        10HD                    # Anunc
                10FD                    # Anunc
                100HD               # Anuncia 100BA
                100FD               # Anuncia 100BA
        Puede concatenar estas opciones para an
orden que van los pulsos en la ráfaga 10HD 10FD
                10HD10FD            # Anuncia 10BAS
                10FD100HD           # Anuncia 10BAS
                10HD100FD           # Anuncia 10BAS
                10HD10FD100HD100FD      # Anunc
                all                 # Anuncia 100FD
ixt11usr1@huffmanKVM:~$
ixt11usr1@huffmanKVM:~$
ixt11usr1@huffmanKVM:~$
ixt11usr1@huffmanKVM:~$
ixt11usr1@huffmanKVM:~$ advertise.sh 10HD10FD
[sudo] password for ixt11usr1:
```

advertise.sh - Notepad

File Edit Format View Help

```
DEVICE="eth1_real"

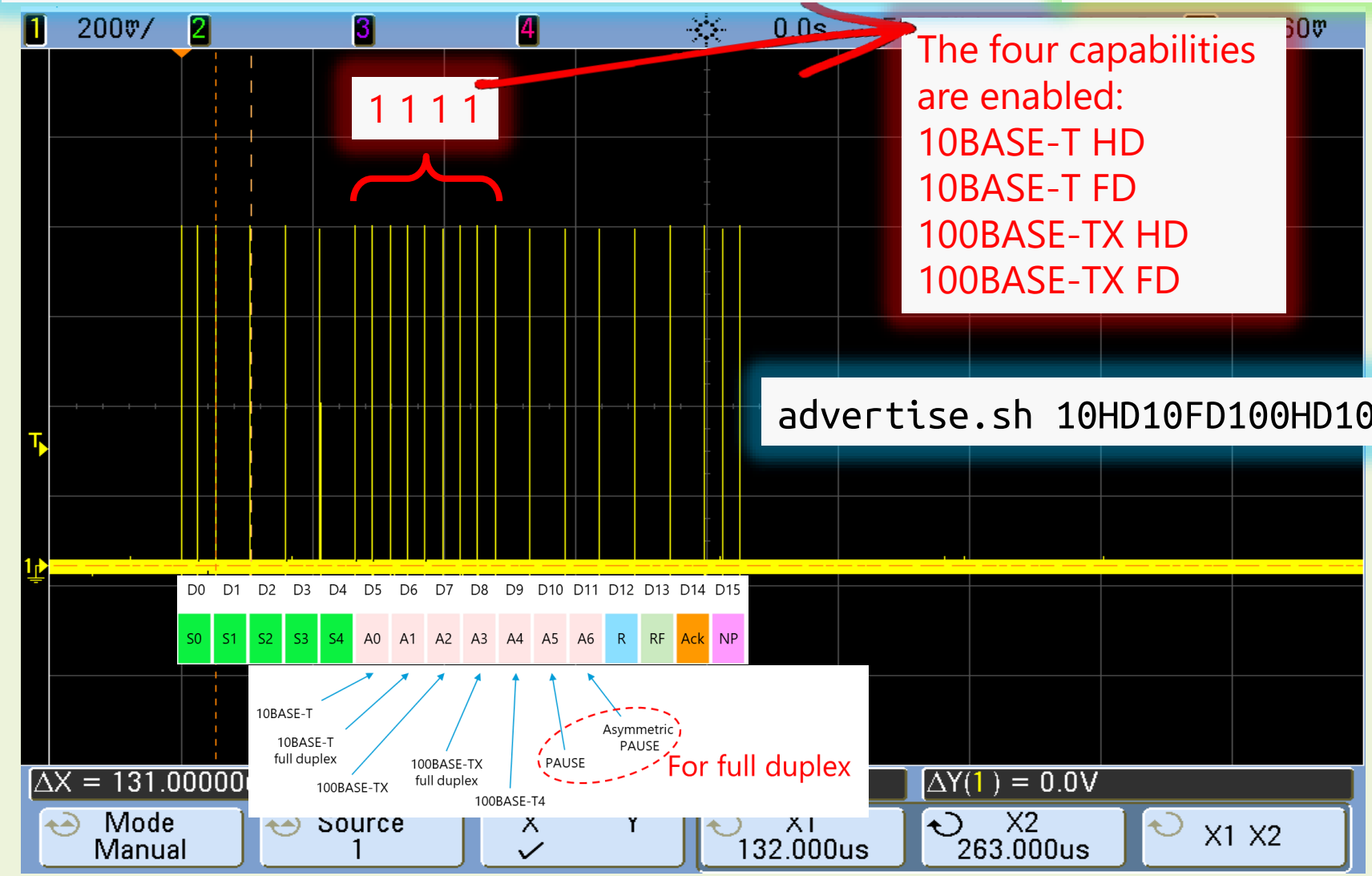
if [[ $1 == "10HD" ]]; then
        sudo ethtool -s $DEVICE advertise 0x1
        exit
fi
if [[ $1 == "10FD" ]]; then
        sudo ethtool -s $DEVICE advertise 0x2
        exit
fi
if [[ $1 == "10HD10FD" ]]; then
        sudo ethtool -s $DEVICE advertise 0x3
        exit
fi
if [[ $1 == "100HD" ]]; then
```

We want to change the **Advertised Link Modes**

Experiment 1
Auto-negotiation Pulses Analysis

Setting of Advertised Link Modes

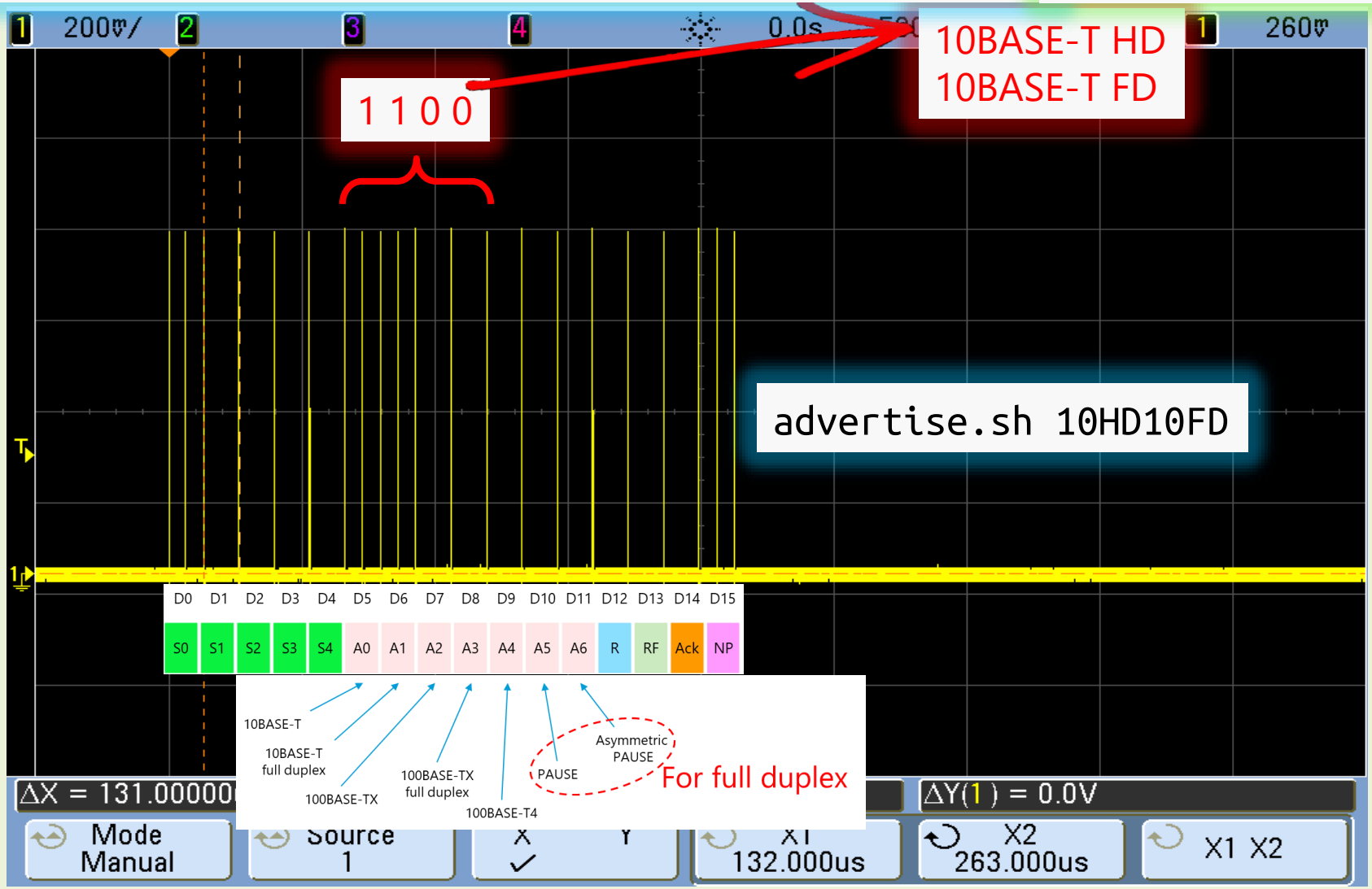
FLP (all capabilities)



Experiment 1
Auto-negotiation Pulses Analysis

Setting of Advertised Link Modes

FLP (10Mbps only)

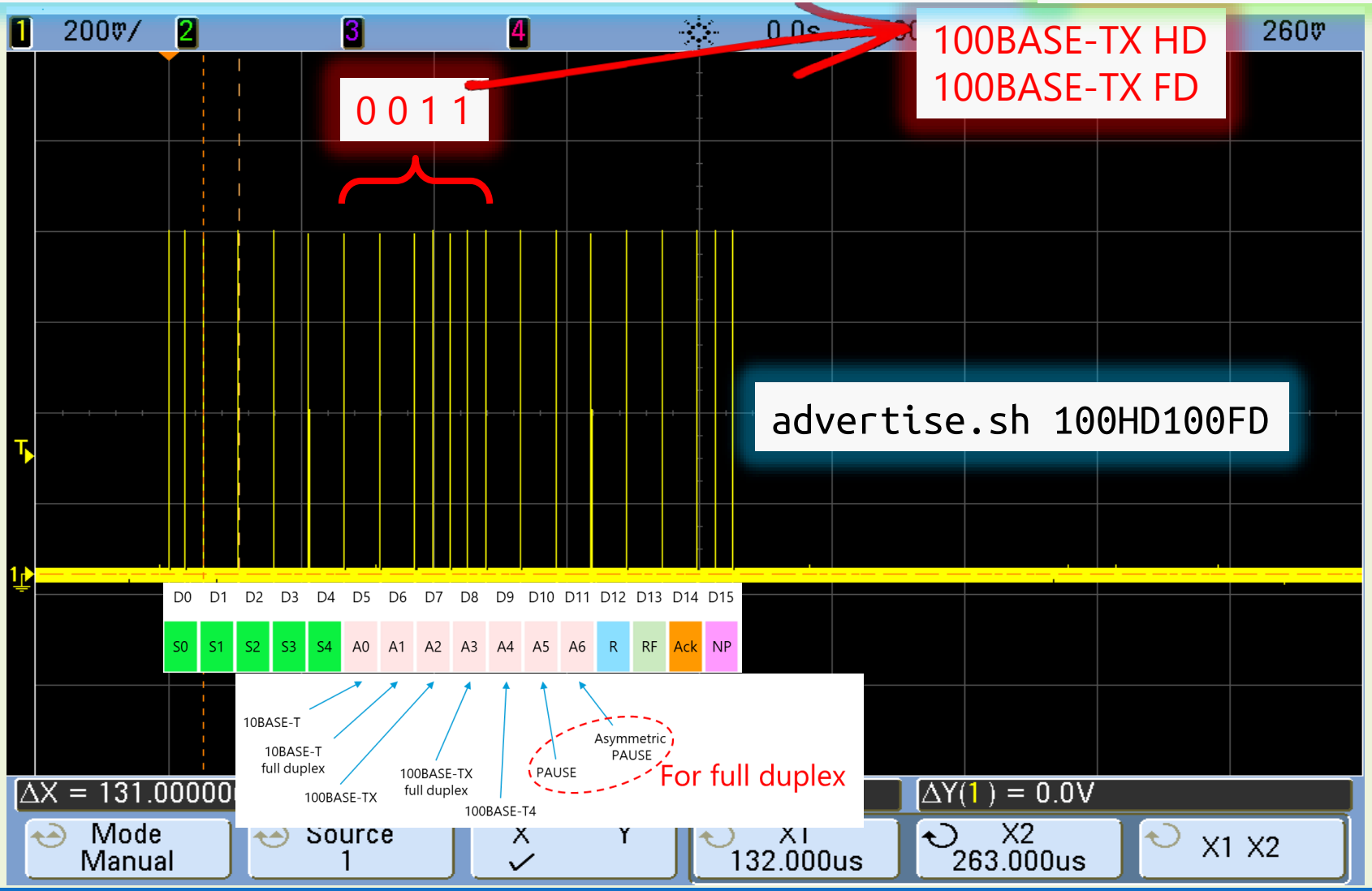


Experiment 1

Auto-negotiation Pulses Analysis

Setting of Advertised Link Modes

FLP (100Mbps only)



Experiment 1
Auto-negotiation Pulses Analysis

E01

Setting of Advertised Link Modes



Show the FLP Burst ✓

Identify the Link Codeword ✓

Identify the Advertised Link Modes ✓

Lab Exercises
Experiment 2

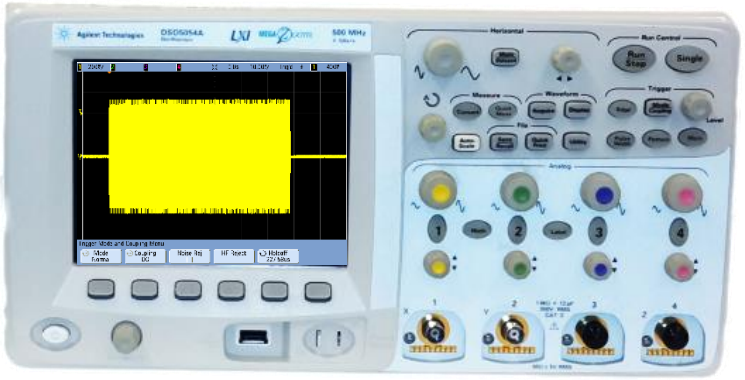
10BASE-T Ethernet Frame Analysis

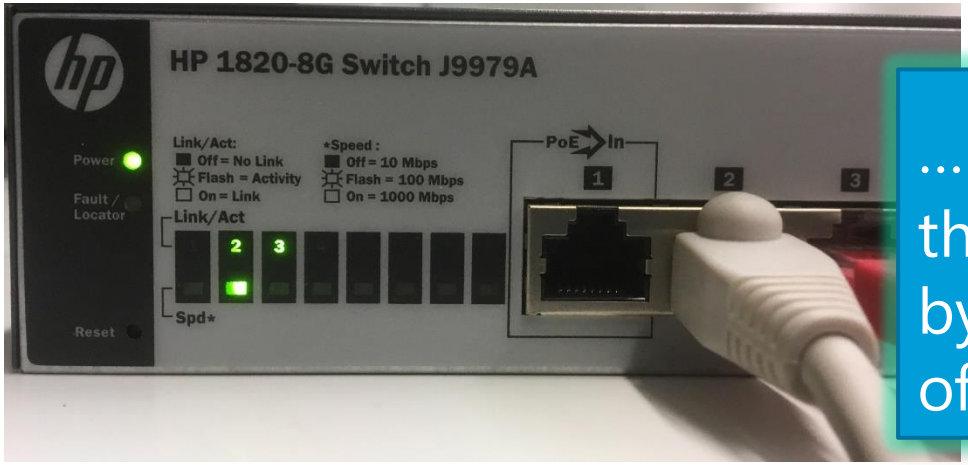


Objectives

You will be able to…

…configure
the oscilloscope to capture 10BASE-T Ethernet frames



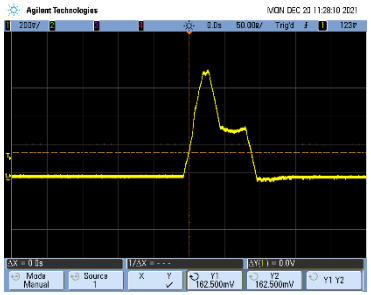


…identify
the link mode of operation by interpreting the LEDs of the switch

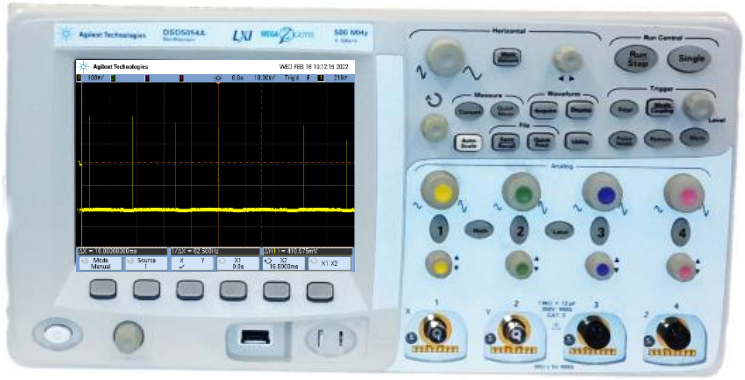
Objectives

You will be able to…

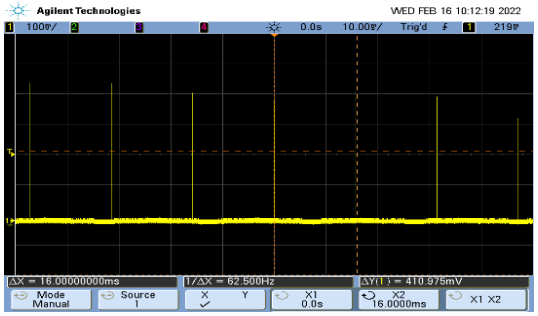
…identify
the link operating mode
according to the Link Signaling



when
link is UP but IDLE



…identify
Normal Link Pulses (NLP)

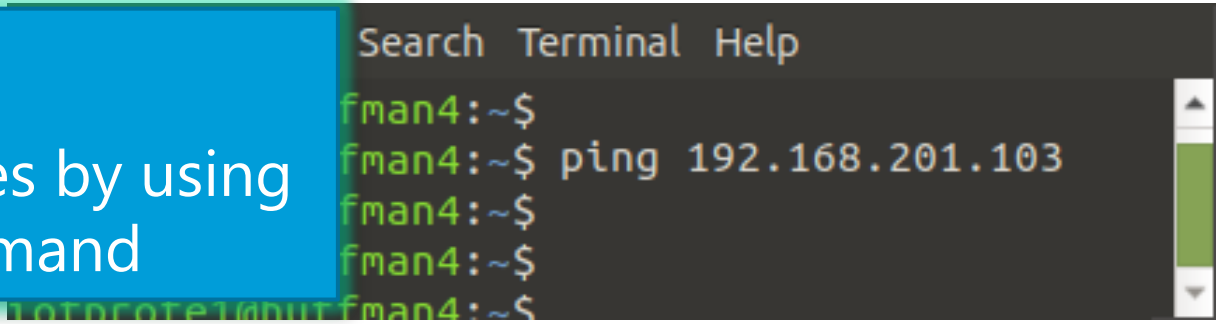


Objectives

You will be able to…

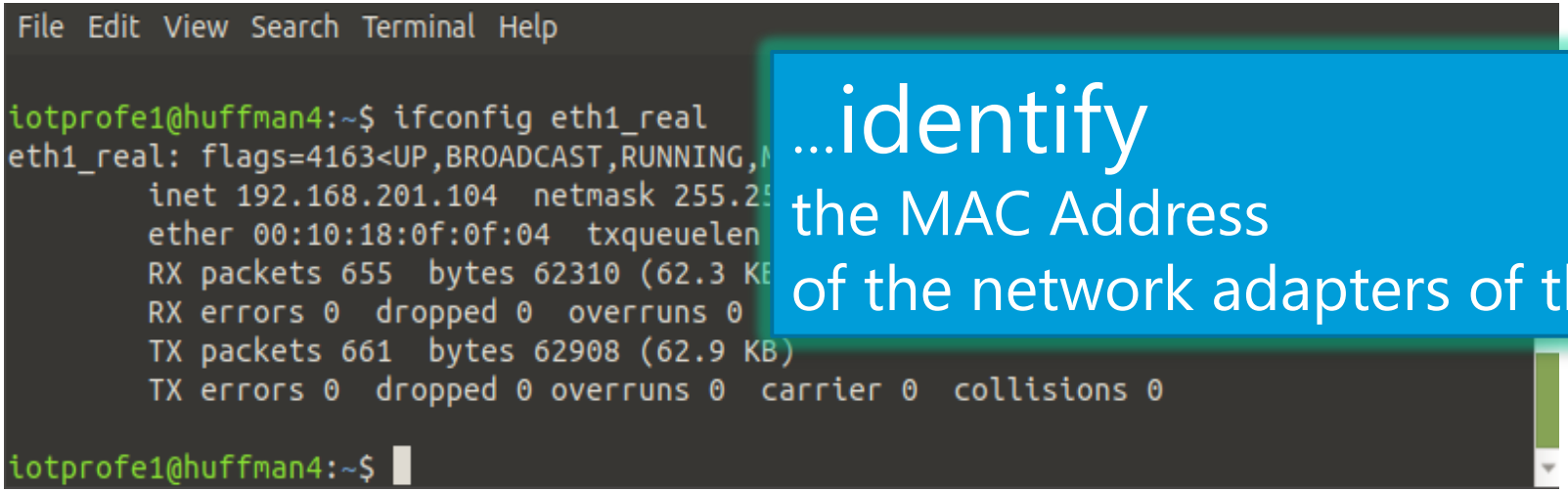
…send

Ethernet frames by using the `ping` command



…identify

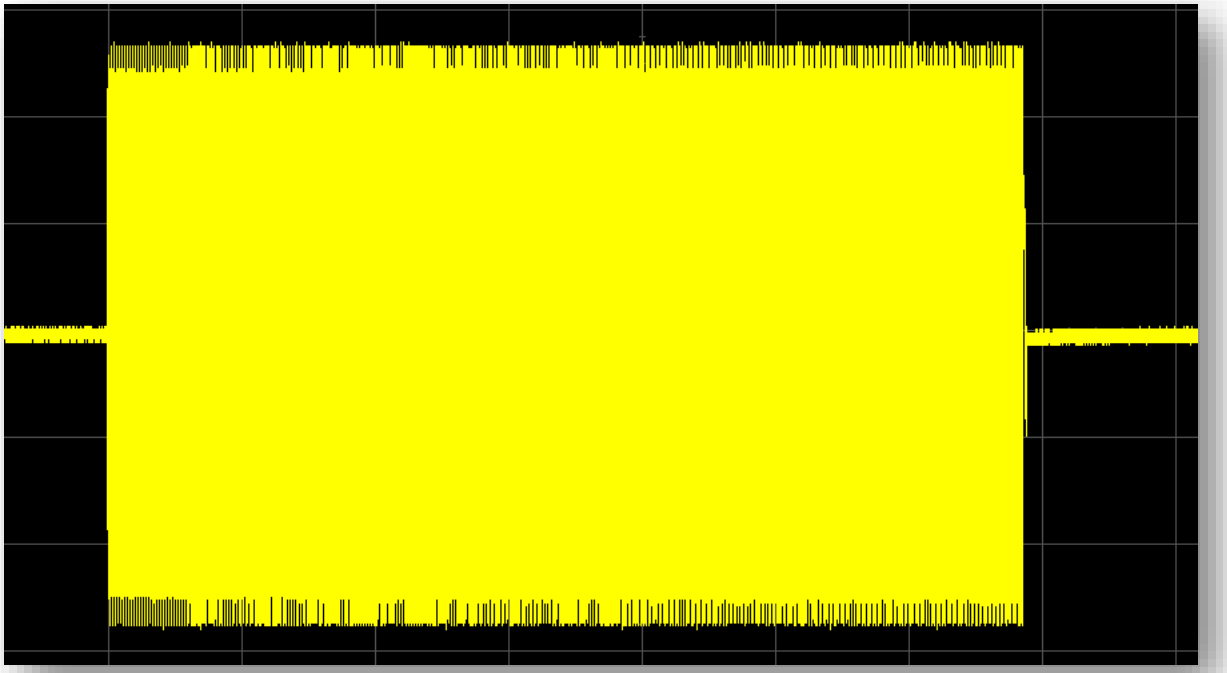
the MAC Address of the network adapters of the PC



Objectives

You will be able to…

…analyze
the capture of an Ethernet frame



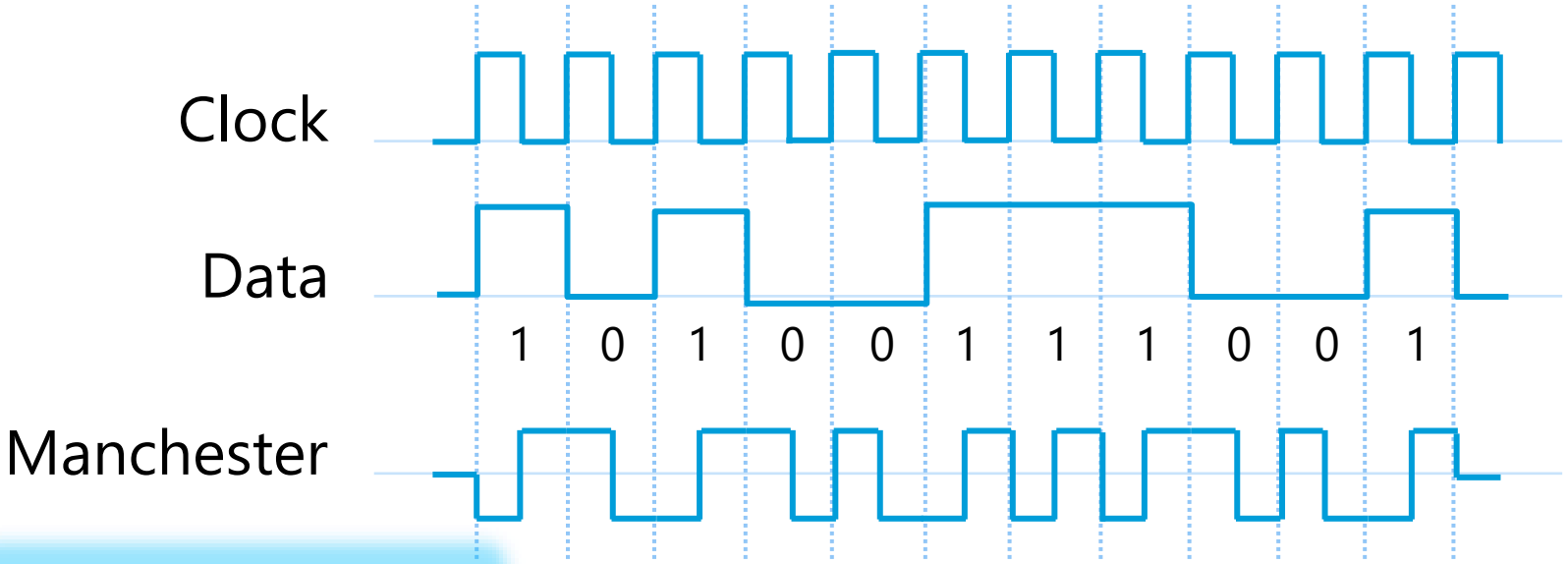
Experiment 2

10BASE-T Ethernet Frame Analysis

Manchester
Coding



Auto-synchronized



Doubles
the required BW

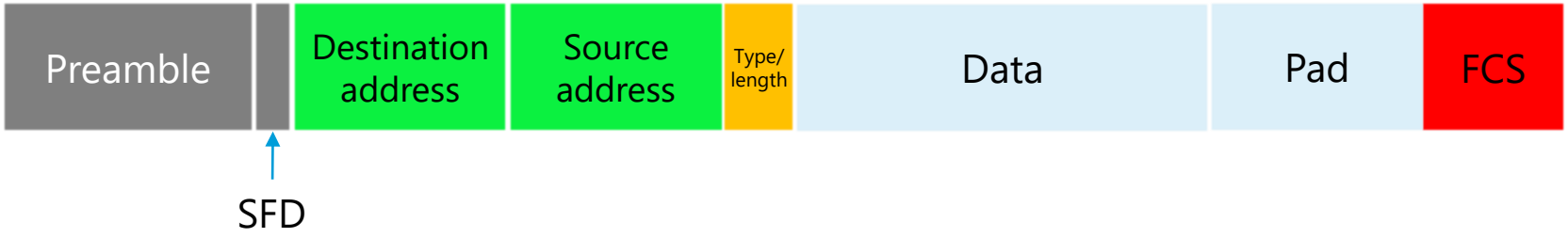




Experiment 2

10BASE-T Ethernet Frame Analysis

Ethernet
Frame





LSB first

Experiment 2

10BASE-T Ethernet Frame Analysis

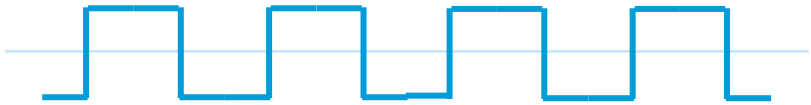
Ethernet
Frame

7 bytes

Preamble

This byte is repeated 7 times

1 0 1 0 1 0 1 0



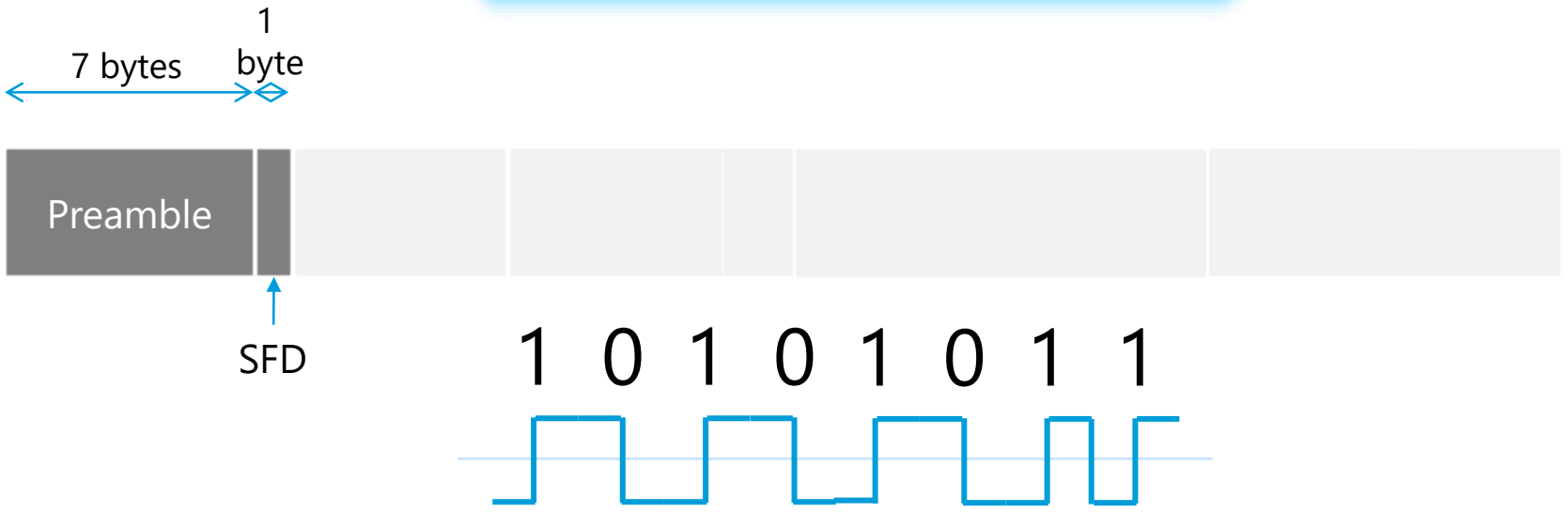
Synchronization of receiver's clock

Experiment 2

10BASE-T Ethernet Frame Analysis

Ethernet
Frame

Start of Frame Delimiter

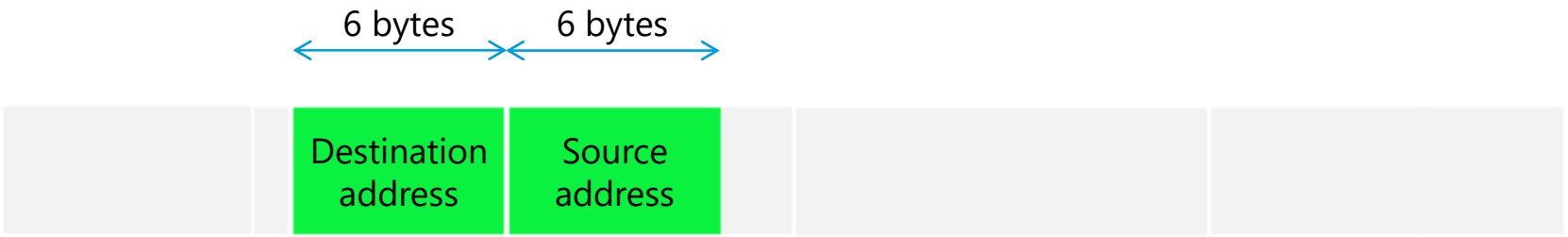


Indicates the start of the Ethernet Frame

Experiment 2

10BASE-T Ethernet Frame Analysis

Ethernet
Frame



Destination and Source Address

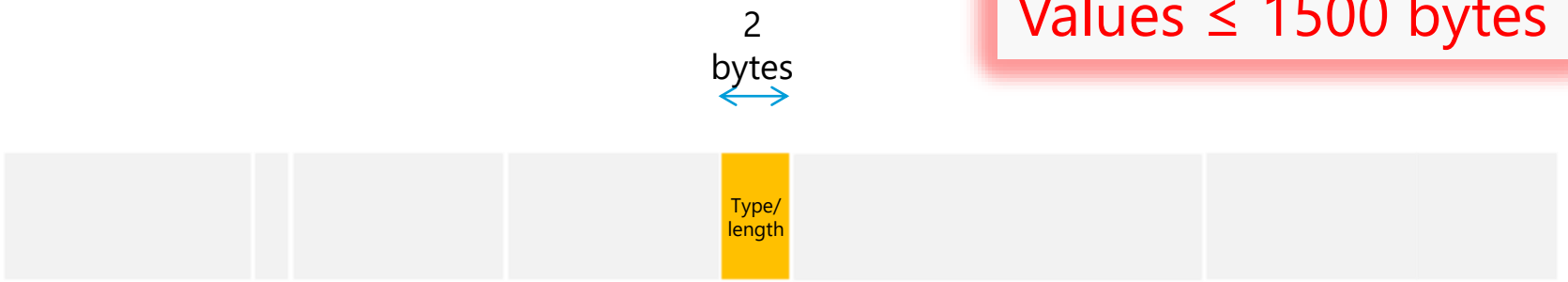
Experiment 2

10BASE-T Ethernet Frame Analysis

Ethernet
Frame

Length of the payload

Values ≤ 1500 bytes



Type of the encapsulated protocol in the payload

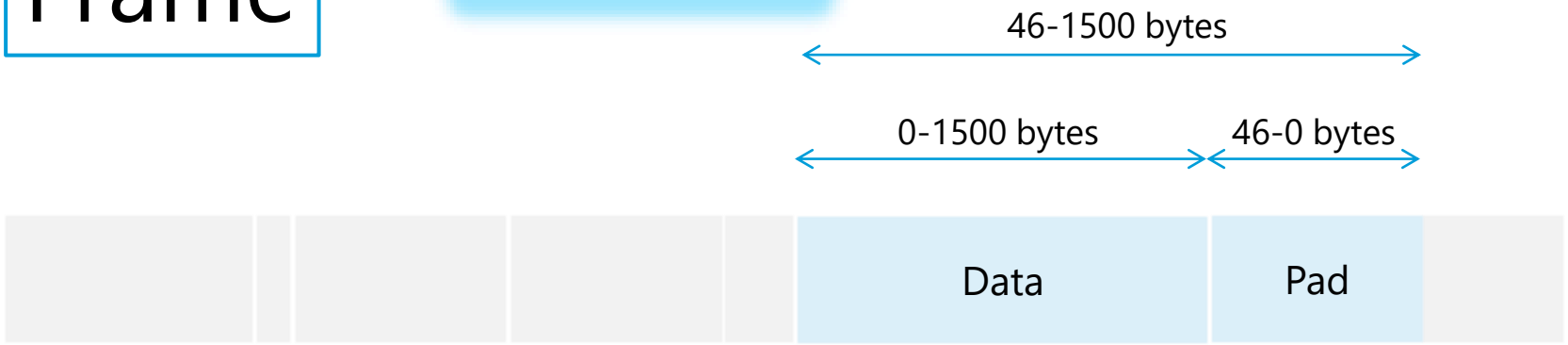
Values ≥ 1536 bytes

Experiment 2

10BASE-T Ethernet Frame Analysis

Ethernet
Frame

Payload



It is used
to guarantee...

To detect collisions

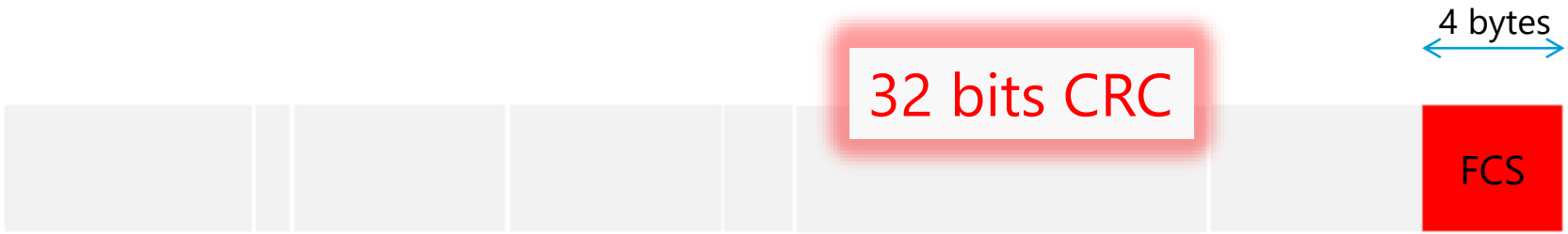
Payload minimum length = 46 bytes

Experiment 2

10BASE-T Ethernet Frame Analysis

Ethernet
Frame

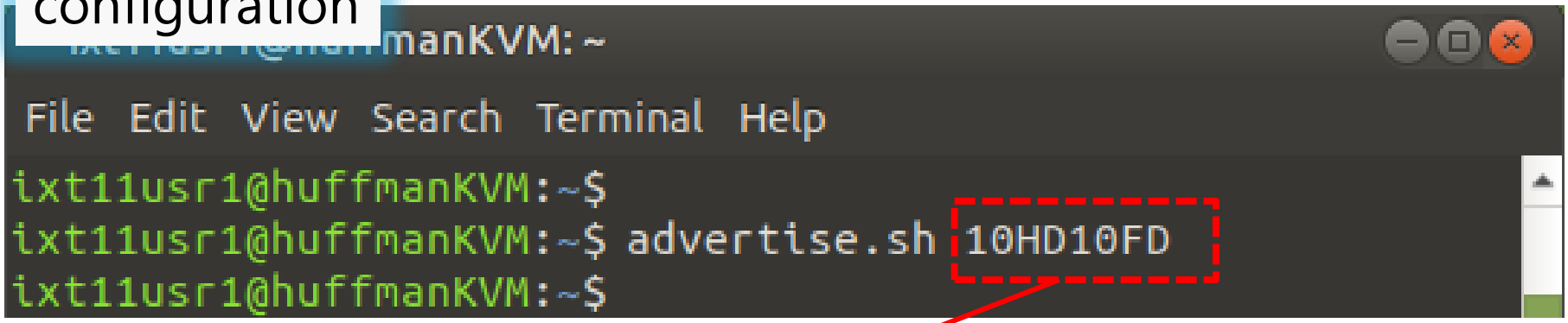
Frame Check Sequence



Experiment 2

10BASE-T Ethernet Frame Analysis

PC Port
configuration



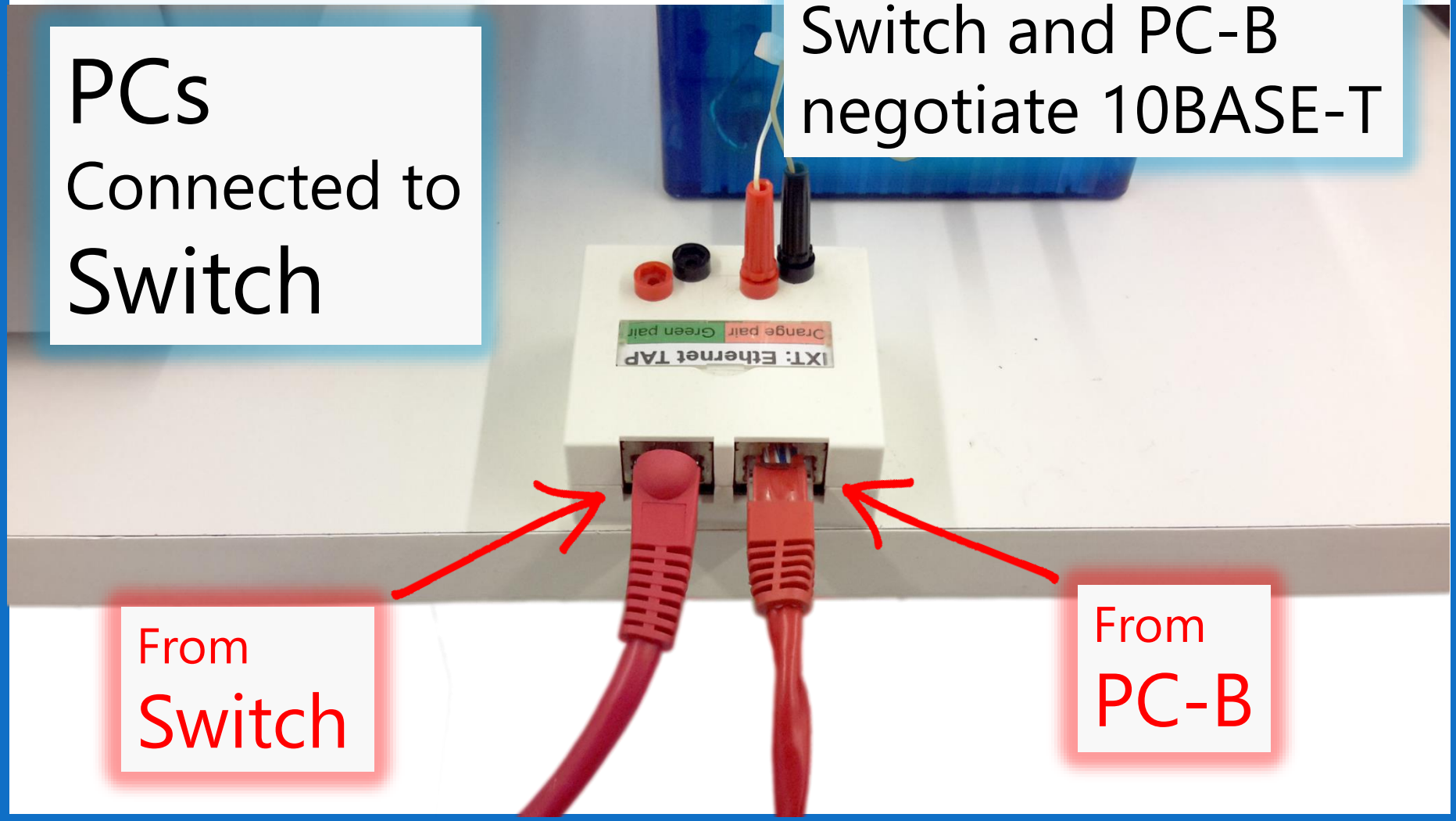
10BASE-T

Make sure
it is just 10Mbps

In this case,
it is HD and FD

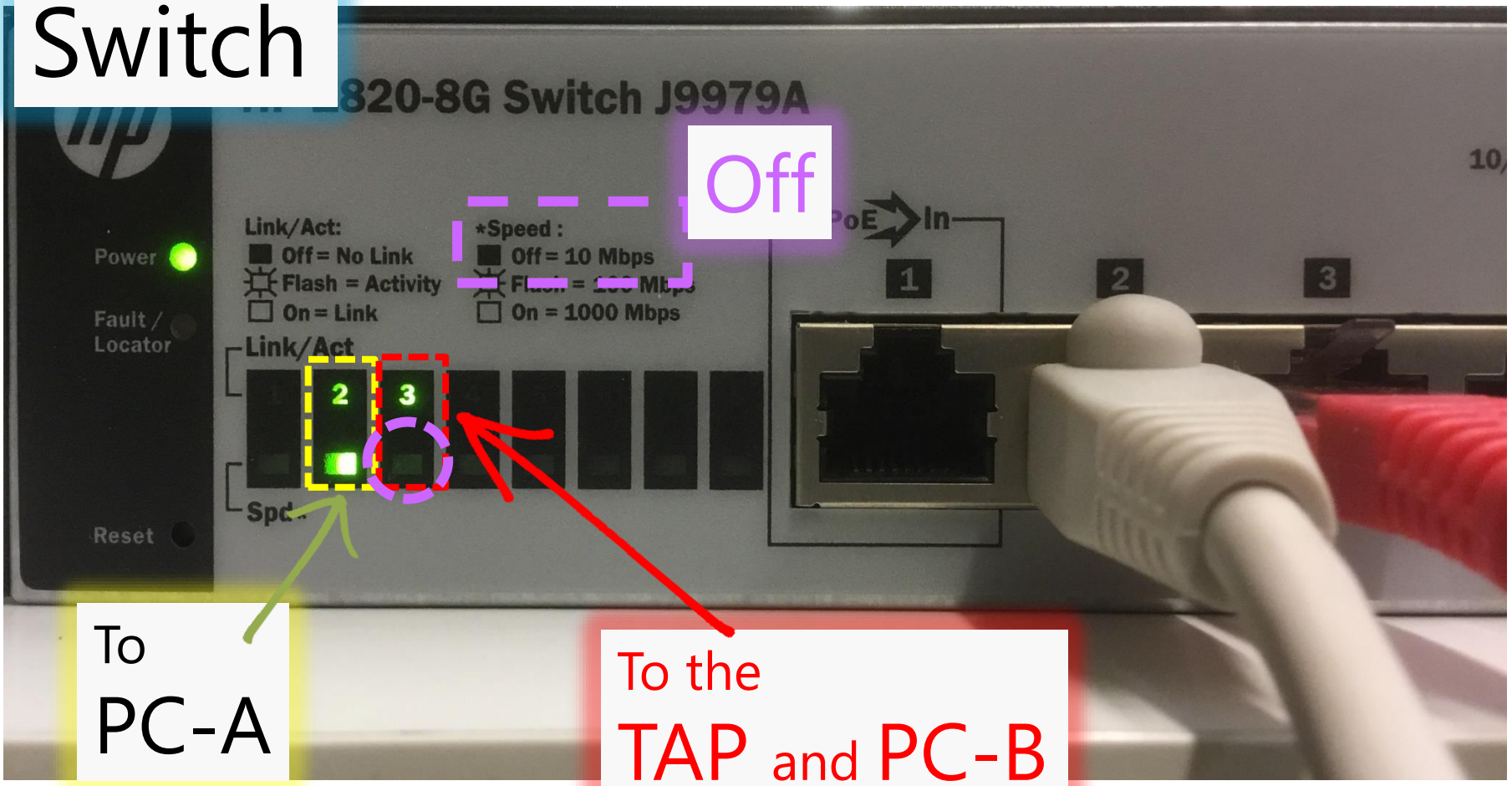
Experiment 2

10BASE-T Ethernet Frame Analysis



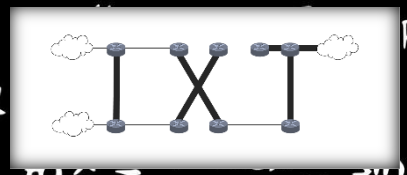
Experiment 2

10BASE-T Ethernet Frame Analysis



Exercises

Experiment 2



10BASE-T Ethernet Frame Analysis

Ethernet
Link Integrity

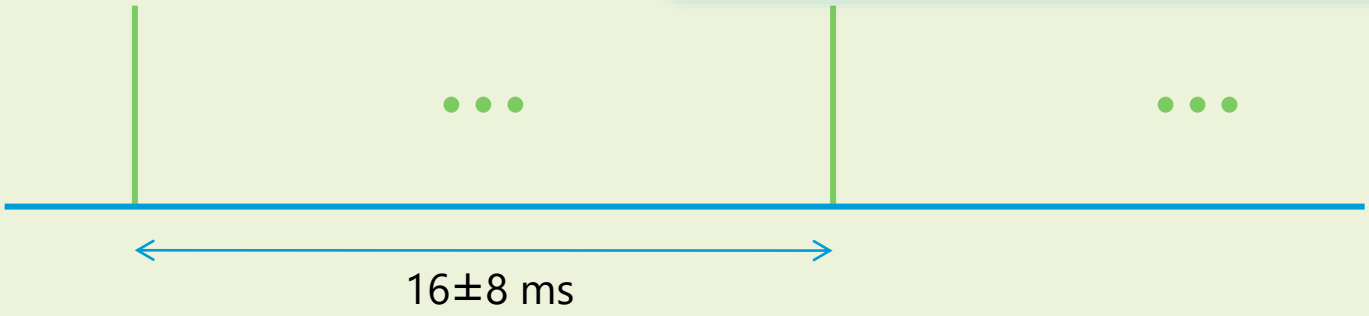


Normal Link Pulse

Transmitted during IDLE periods...

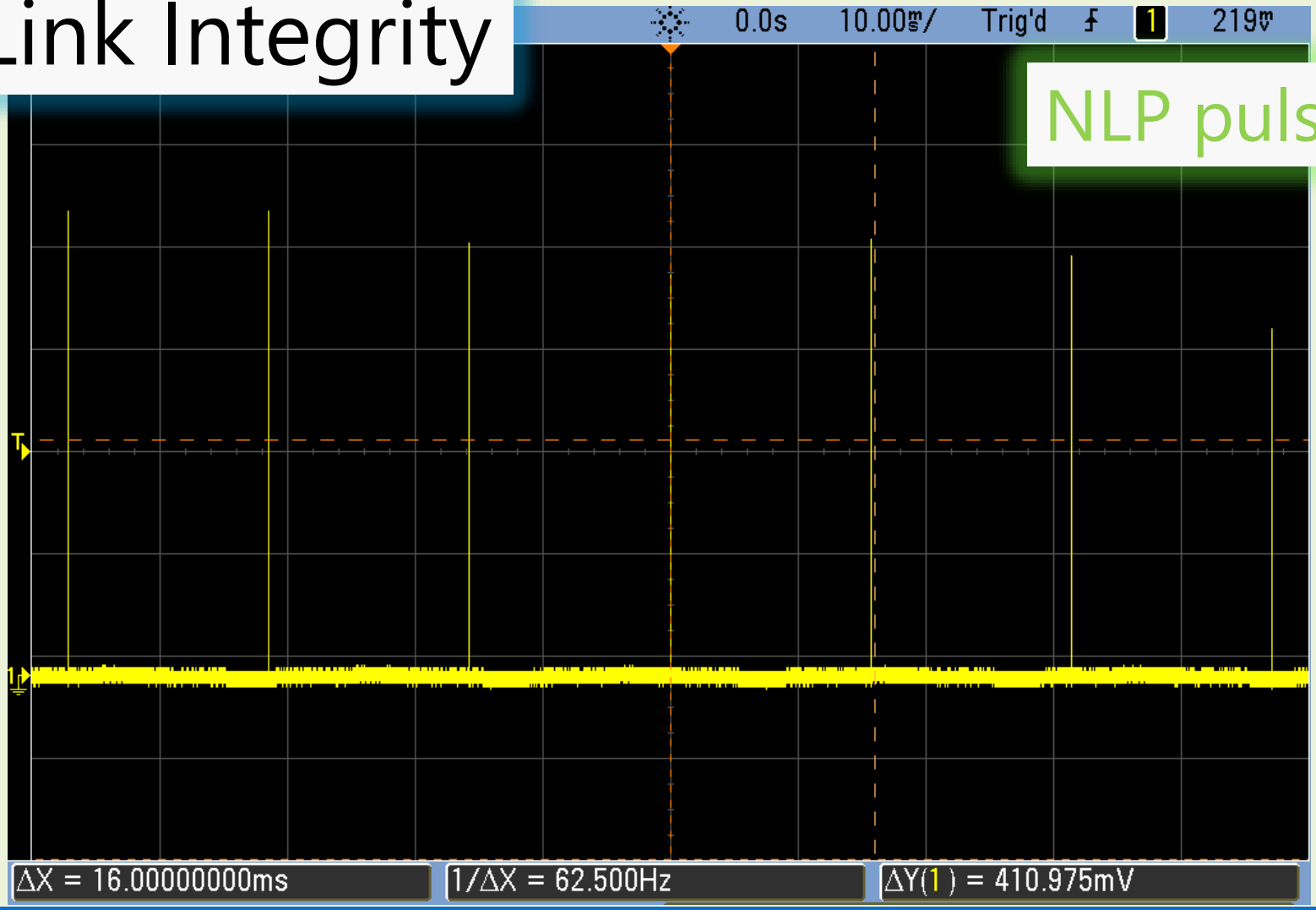
...to determine if a link is successfully established

NLPs



Ethernet

Link Integrity



NLP pulses

Now, send frames



Ping command from PC-B to PC-A

```
iotprofe1@huffman4:~$
iotprofe1@huffman4:~$ ping 192.168.201.103
iotprofe1@huffman4:~$
iotprofe1@huffman4:~
iotprofe1@huffman4:~
```

@IP

.1yy

yy is according to PC-A @IP

Experiment 2

10BASE-T Ethernet Frame Analysis

Capture an Ethernet Frame

- Identify the Ethernet frame
- Identify the waveform of bits 1 and 0
- Identify the frequency of the signal
- Identify the SFD field
- Decodify the destination address
- Identify pros and cons of Manchester Coding

How to know your Ethernet addresses...

MAC address

```
File  Edit  View  Search  Terminal  Help

iotprofe1@huffman4:~$ ifconfig eth1_real
eth1_real: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.201.104  netmask 255.255.255.0  broadcast 192.168.201.255
        ether 00:10:18:0f:0f:04  txqueuelen 1000  (Ethernet)
        RX packets 655  bytes 62310 (62.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 661  bytes 62908 (62.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

iotprofe1@huffman4:~$
```

How to know your Ethernet addresses...

```
iotprofe1@huffman4:~$ ifconfig
eth0_real: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.30.104  netmask 255.255.255.0  broadcast 10.0.30.255
        ether 00:d8:61:98:a4:90  txqueuelen 1000  (Ethernet)
        RX packets 20220  bytes 24067516 (24.0 MB)
        RX errors 0  dropped 444  overruns 0  frame 0
        TX packets 11225  bytes 2325733 (2.3 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 17  memory 0xb2300000-b2320000

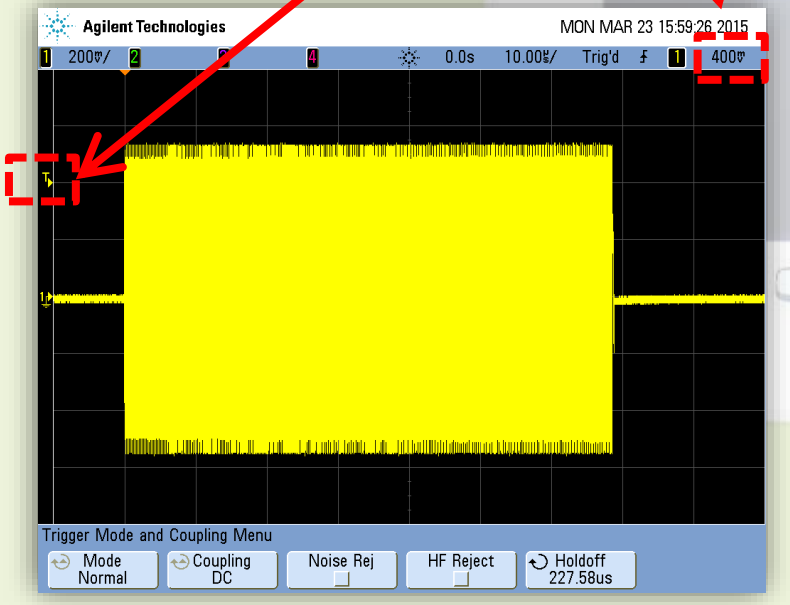
eth1_real: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.201.104  netmask 255.255.255.0  broadcast 192.168.201.255
        ether 00:10:18:0f:0f:04  txqueuelen 1000  (Ethernet)
        RX packets 655  bytes 62310 (62.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 661  bytes 62908 (62.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 570  bytes 61411 (61.4 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 570  bytes 61411 (61.4 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Experiment 2

10BASE-T Ethernet Frame Analysis

Trigger level
~460 mV

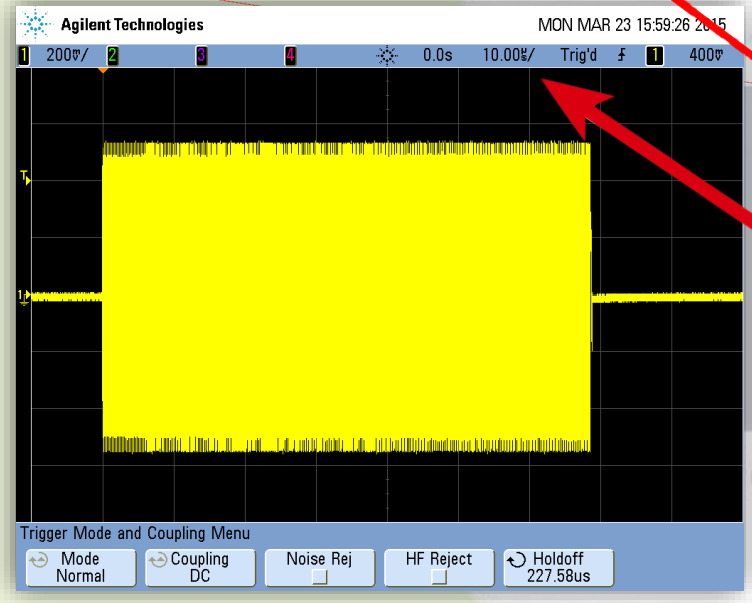




Experiment 2

10BASE-T Ethernet Frame Analysis





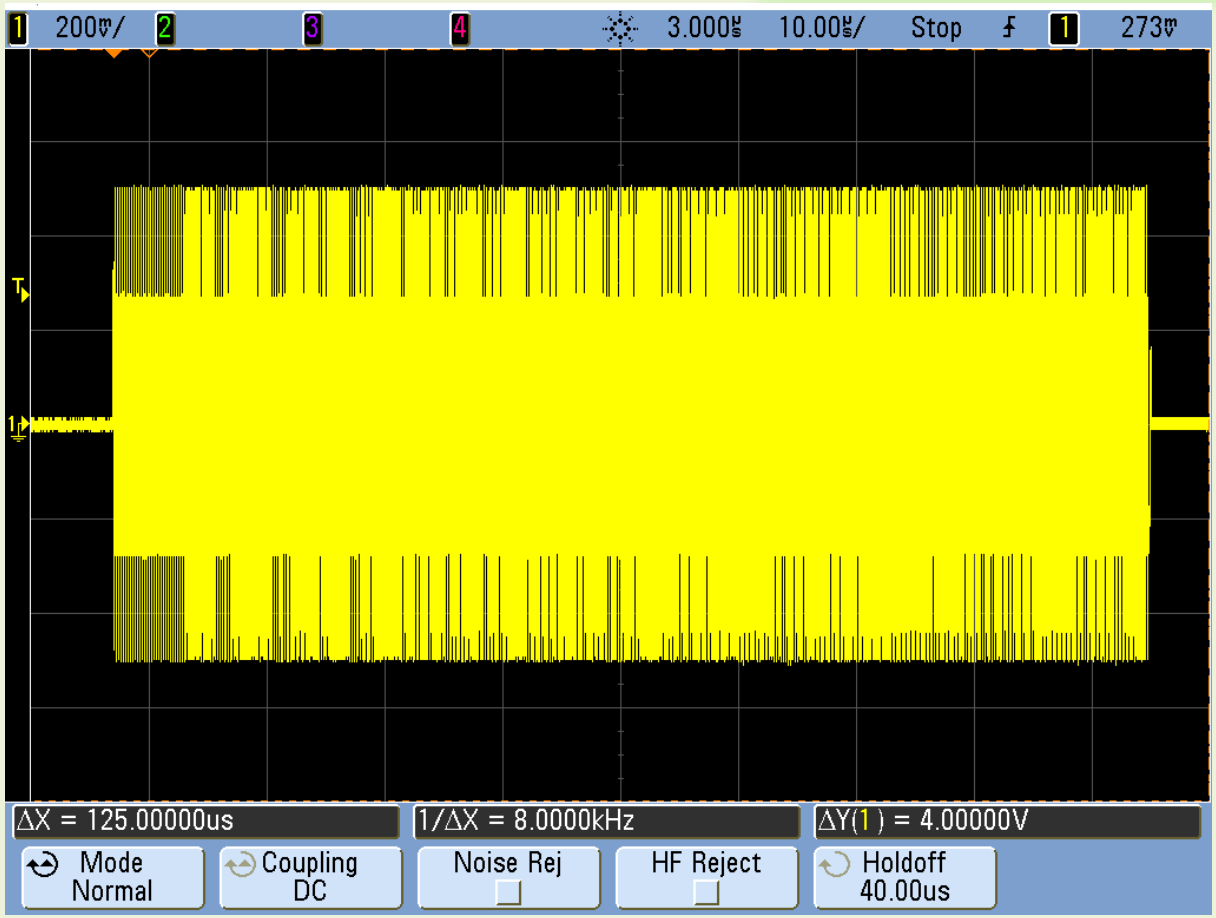
To capture the WHOLE 10BASE-T Ethernet frame

10 μs

Experiment 2

10BASE-T Ethernet Frame Analysis

Identify the whole Ethernet frame



Experiment 2

10BASE-T Ethernet Frame Analysis

E02

Identify the whole Ethernet frame





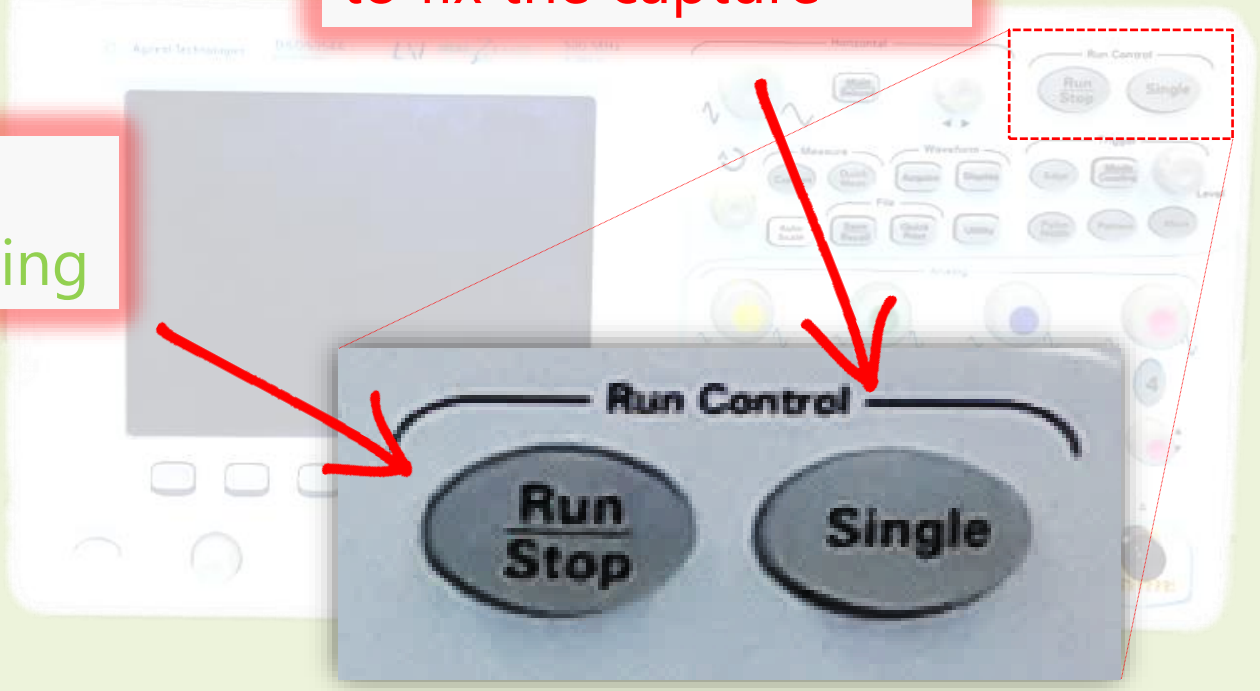
Show the whole Ethernet frame ✓

Experiment 2

10BASE-T Ethernet Frame Analysis

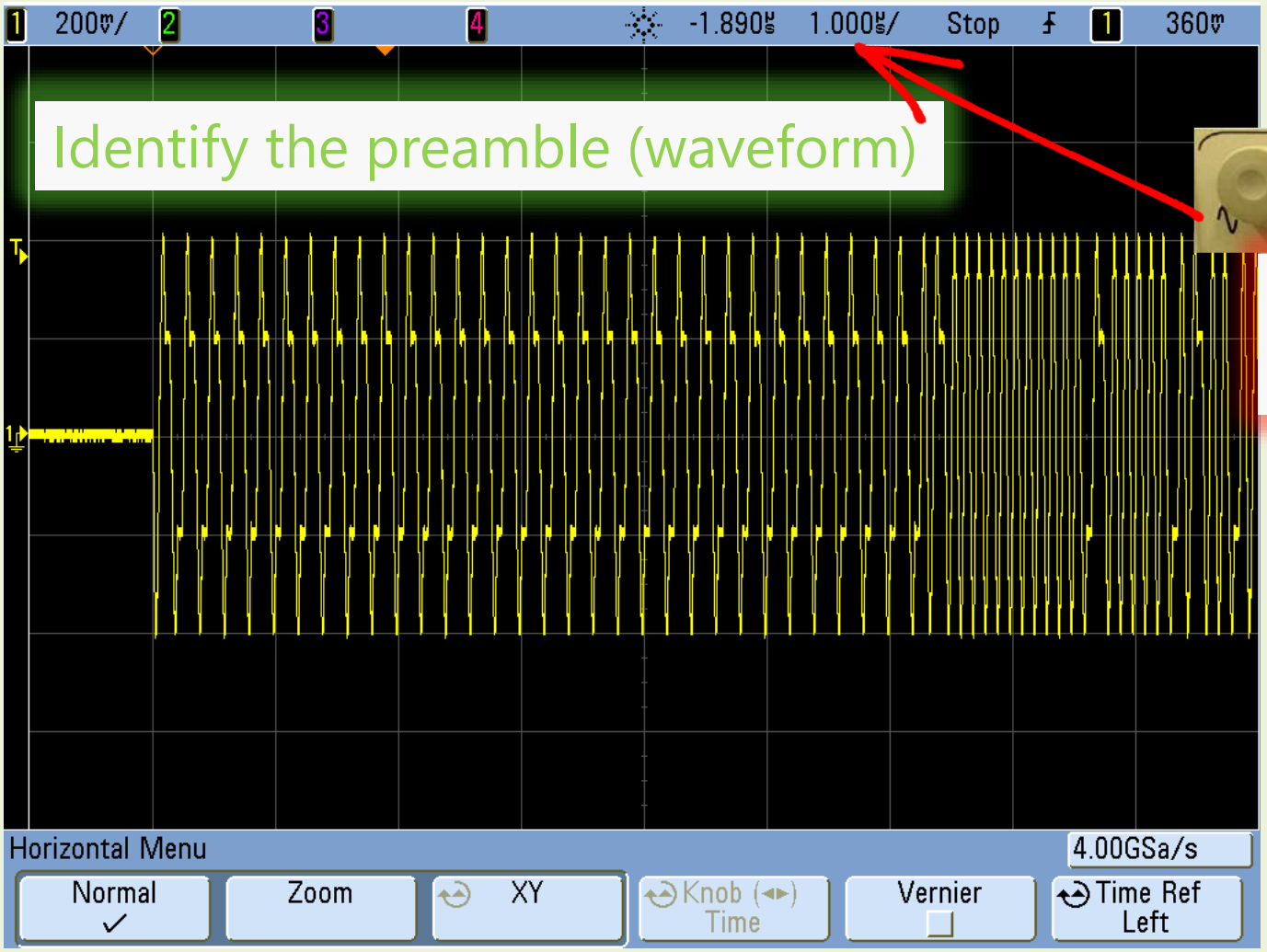
Use single acquisition to fix the capture

RED stopped
GREEN capturing



Experiment 2

10BASE-T Ethernet Frame Analysis

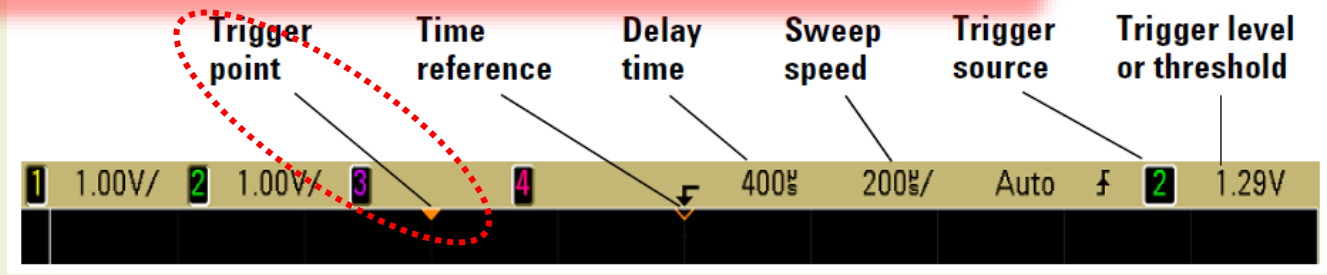


Experiment 2

10BASE-T Ethernet Frame Analysis



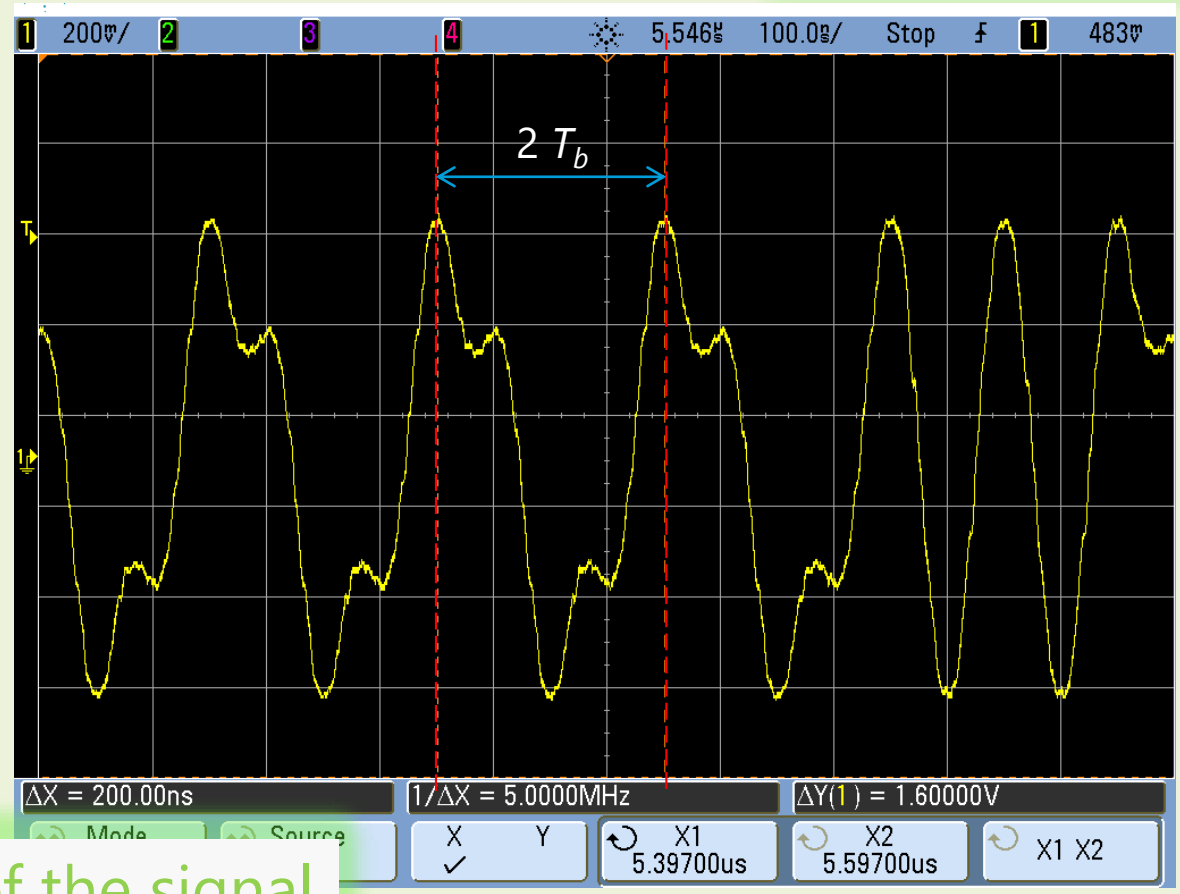
Use the trigger point indicator
to pan through the capture horizontally



Experiment 2

10BASE-T Ethernet Frame Analysis

Frequency of the signal

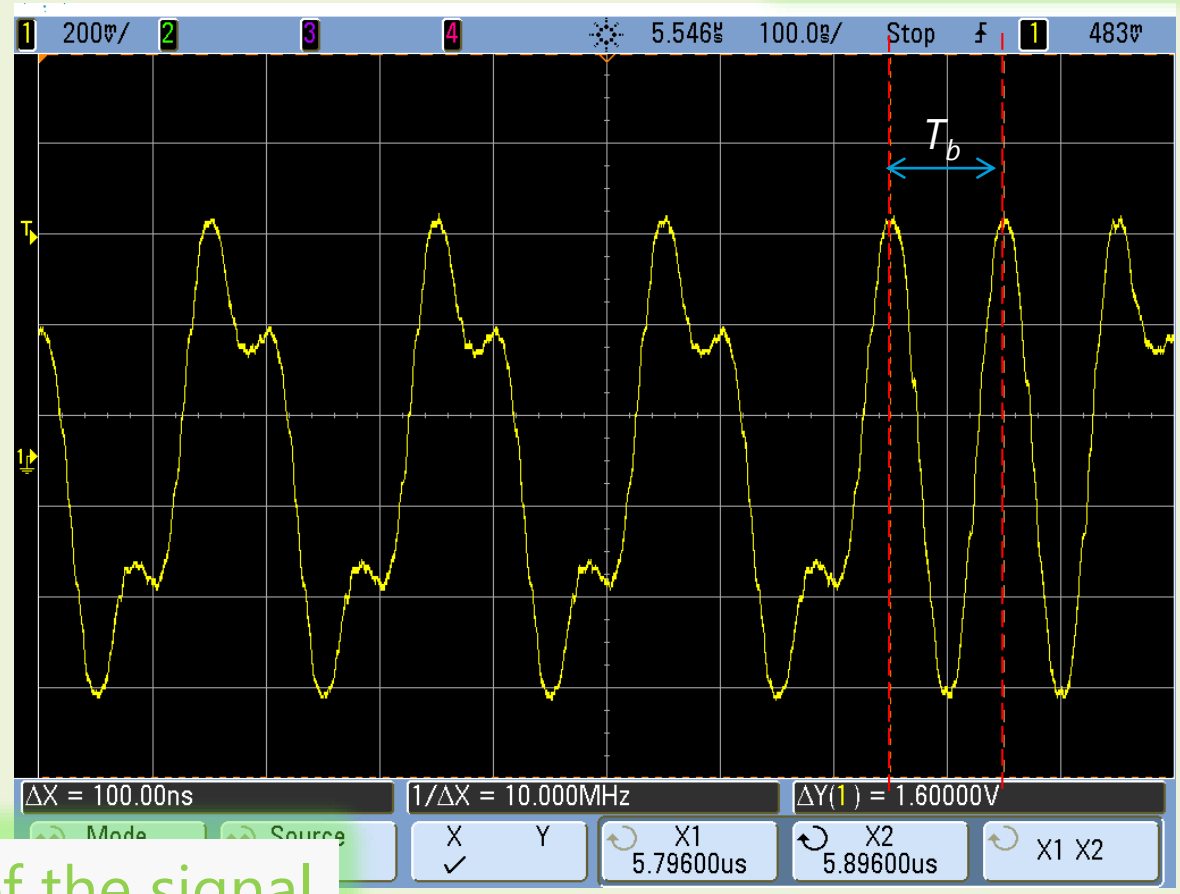


Period of the signal

Experiment 2

10BASE-T Ethernet Frame Analysis

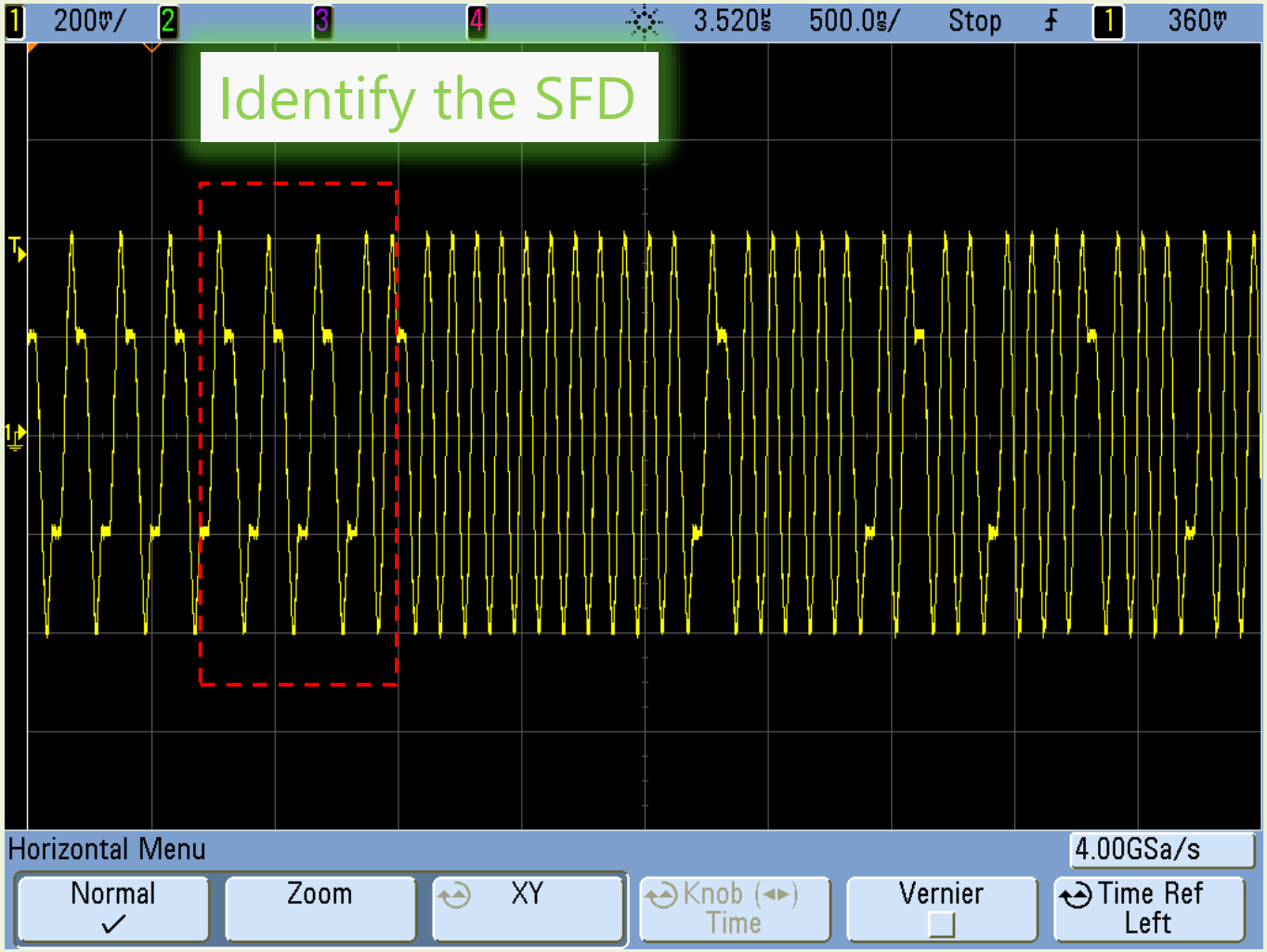
Frequency of the signal



Period of the signal

Experiment 2

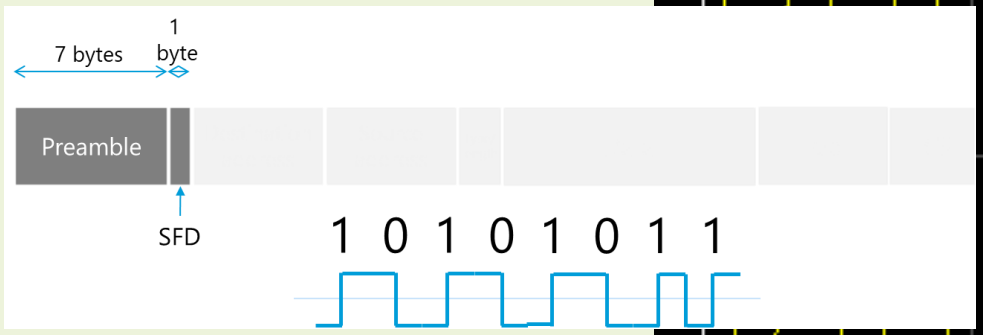
10BASE-T Ethernet Frame Analysis

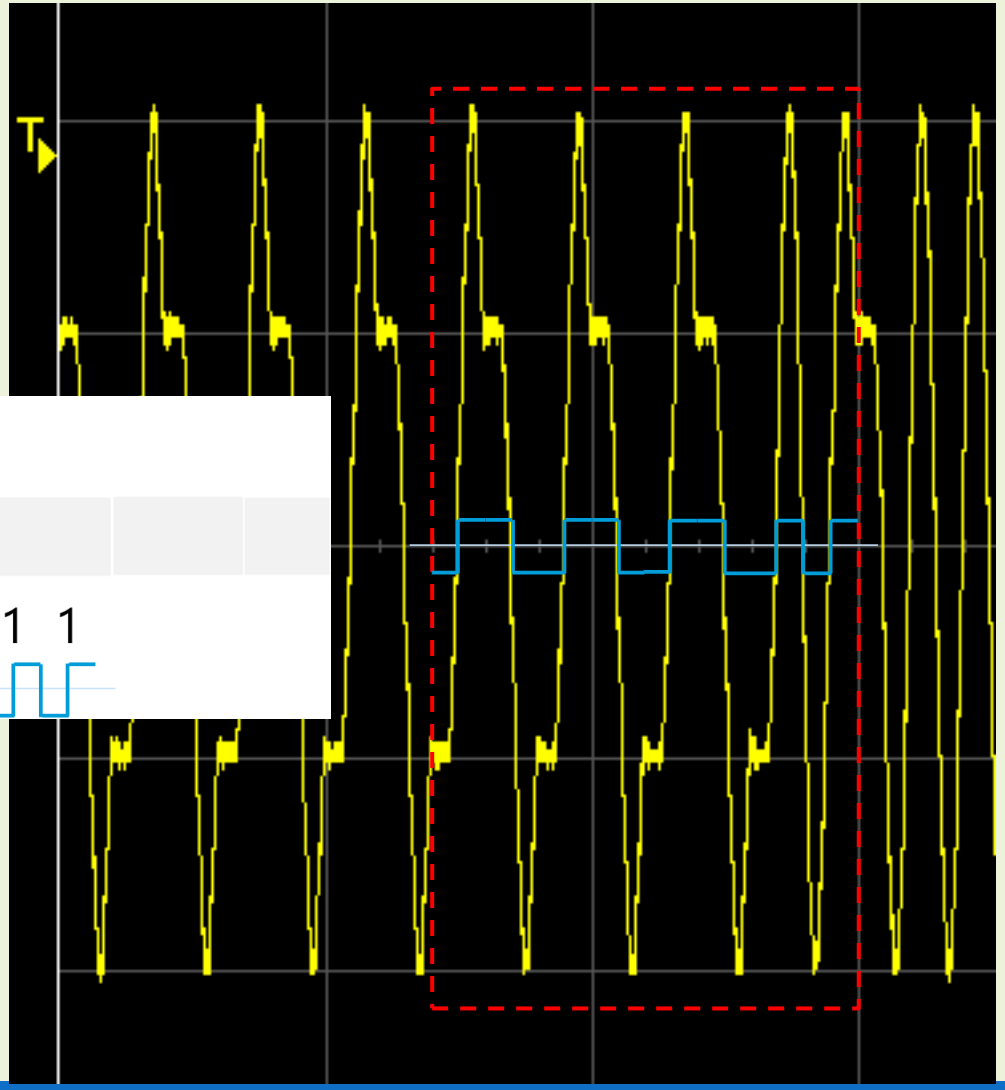


Experiment 2

10BASE-T Ethernet Frame Analysis

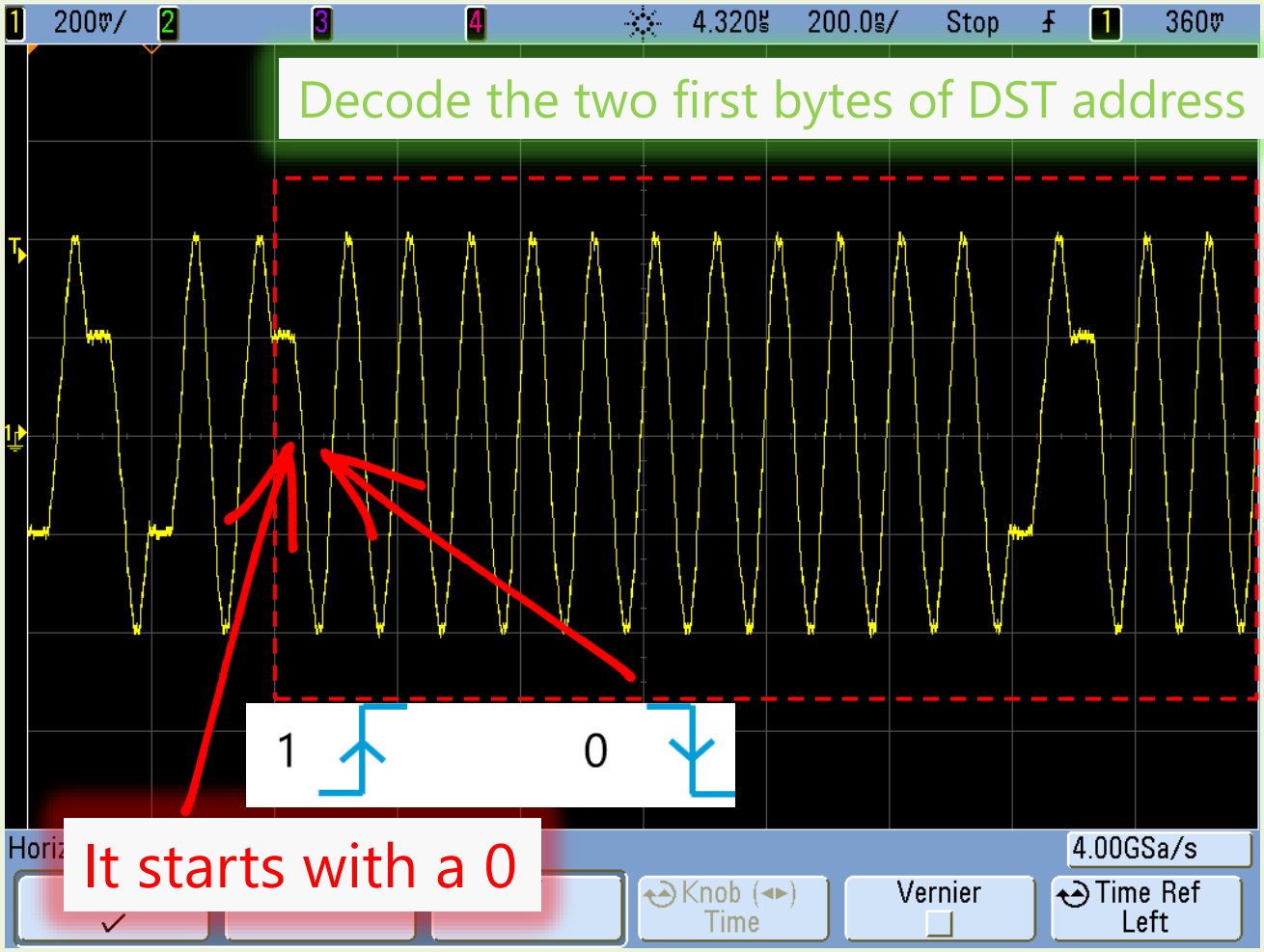
Identify the SFD





Experiment 2

10BASE-T Ethernet Frame Analysis



Experiment 2

10BASE-T Ethernet Frame Analysis

E03

Decode the two first bytes of DST address



Show the first two bytes of the DST address

✓

Lab Exercises
Experiment 3

100BASE-TX Ethernet Signal Analysis



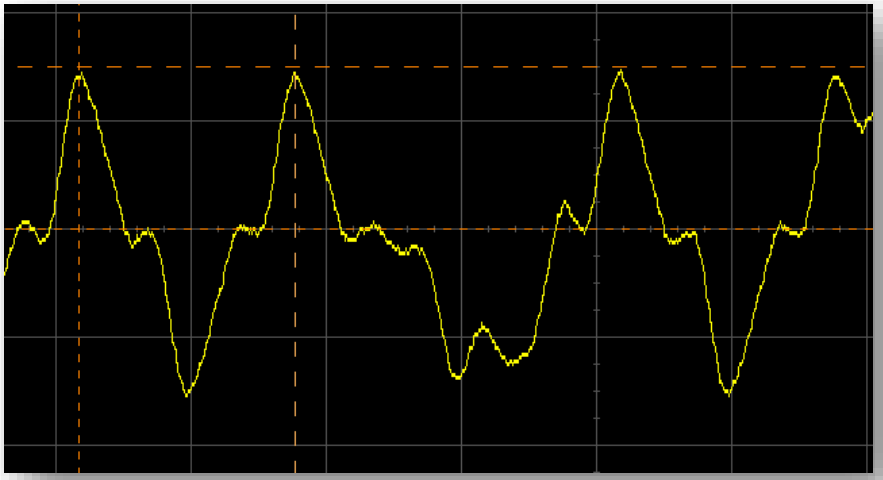
Objectives

You will be able to…



…configure
the oscilloscope to capture an MLT-3 signal

…analyze
a capture of an MLT-3 signal

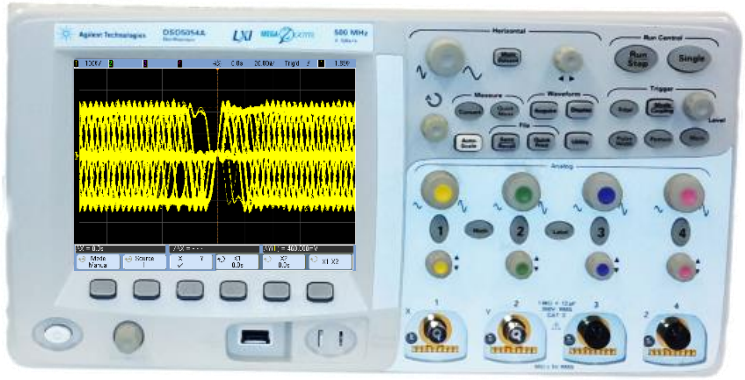


Objectives

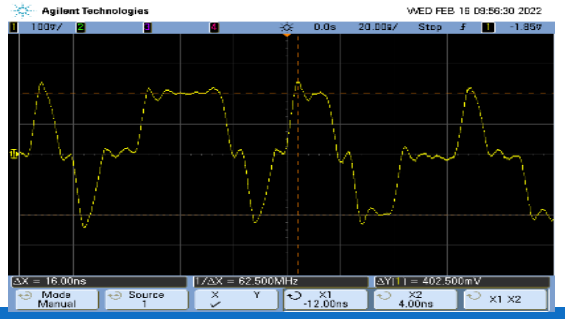
You will be able to…

…identify
the link operating mode
according to the Link Signaling

when
link is UP but IDLE



4B/5B IDLE Character
MLT-3 coding



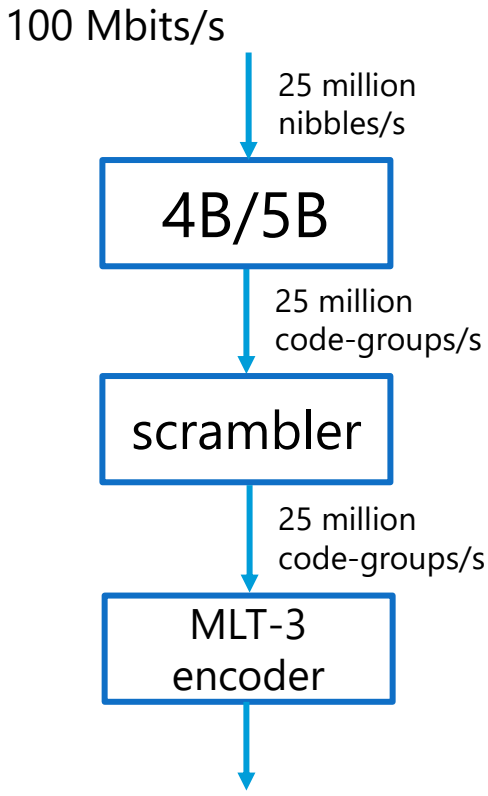
Experiment 3

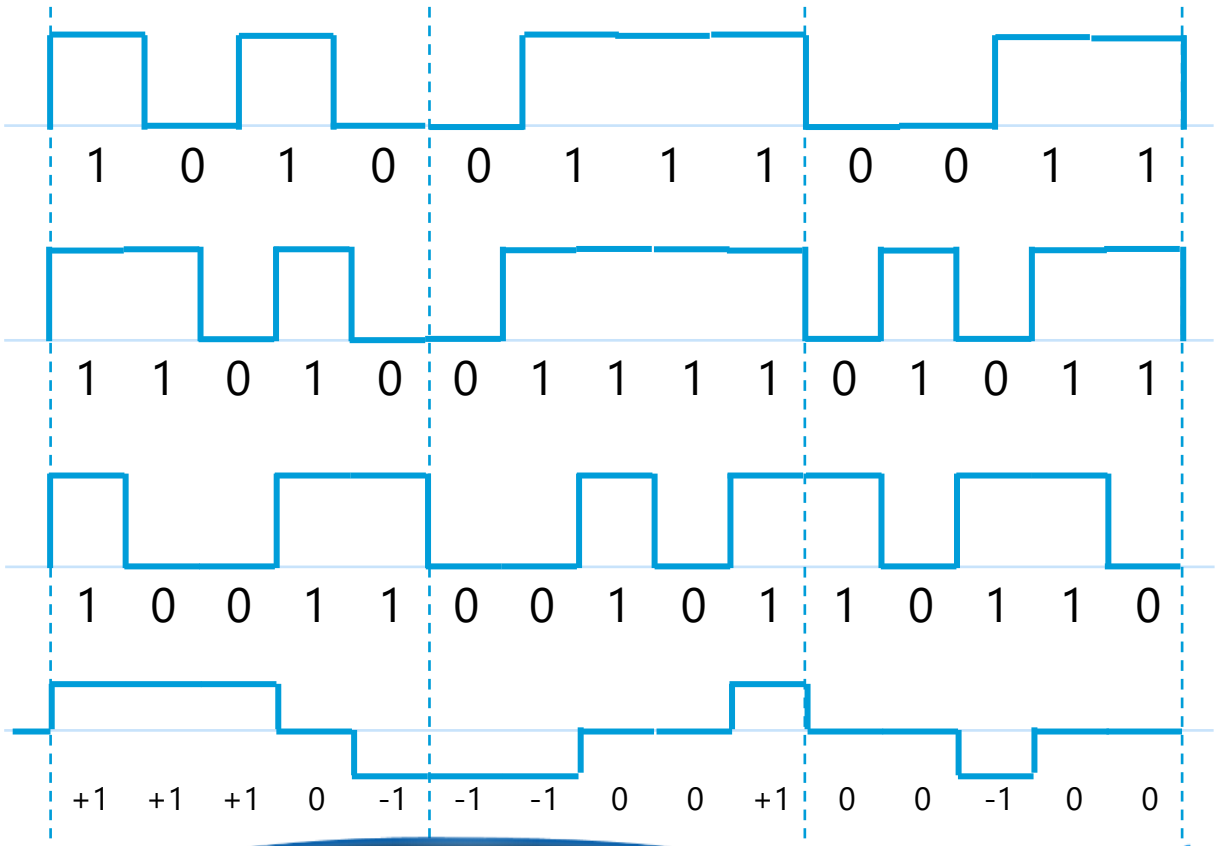
100BASE-TX Ethernet Signal Analysis

MLT-3 coding

Physical Coding

Requires less bandwidth

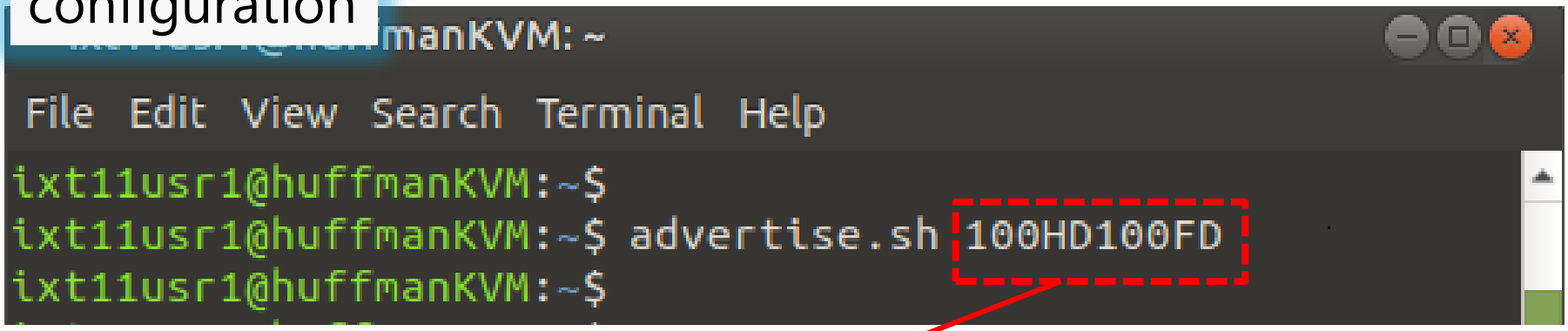




Experiment 3

100BASE-TX Ethernet Signal Analysis

PC Port configuration



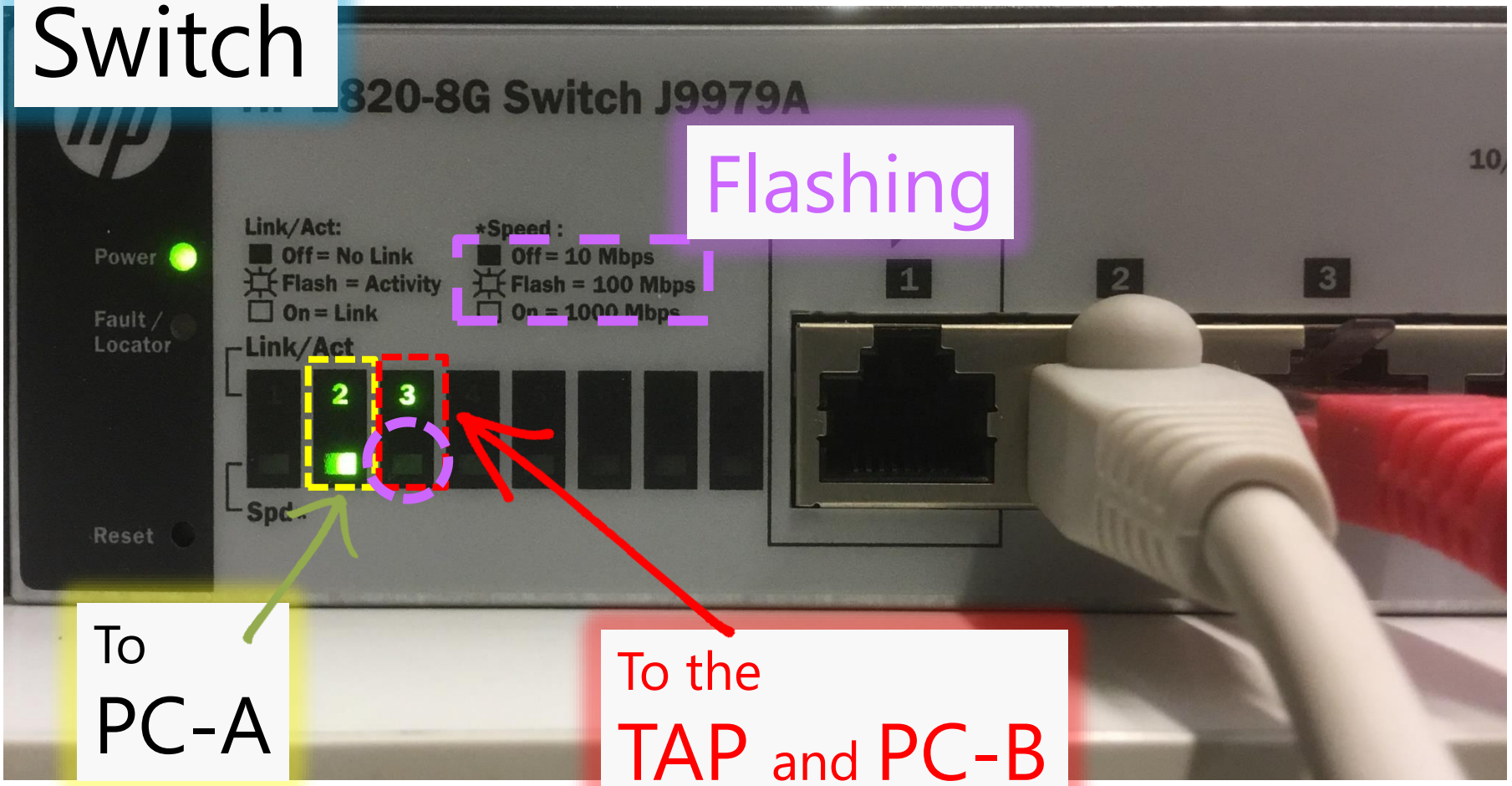
100BASE-T

Make sure it is just 100Mbps

In this case, it is HD and FD

Experiment 3

100BASE-TX Ethernet Signal Analysis

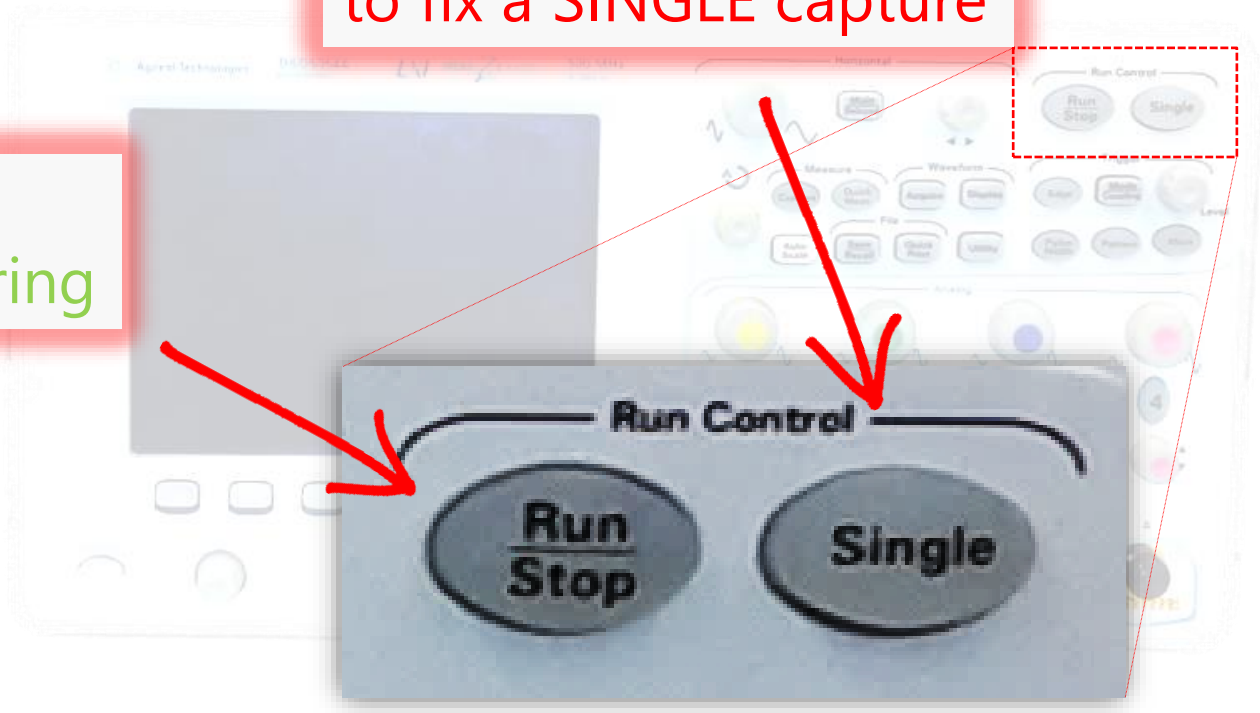


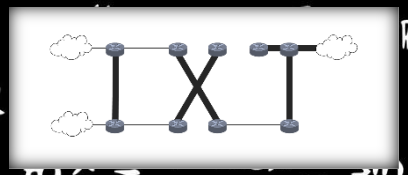
Experiment 3

100BASE-TX Ethernet Signal Analysis

Use single acquisition to fix a SINGLE capture

RED stopped
GREEN capturing





Exercises

Experiment 3

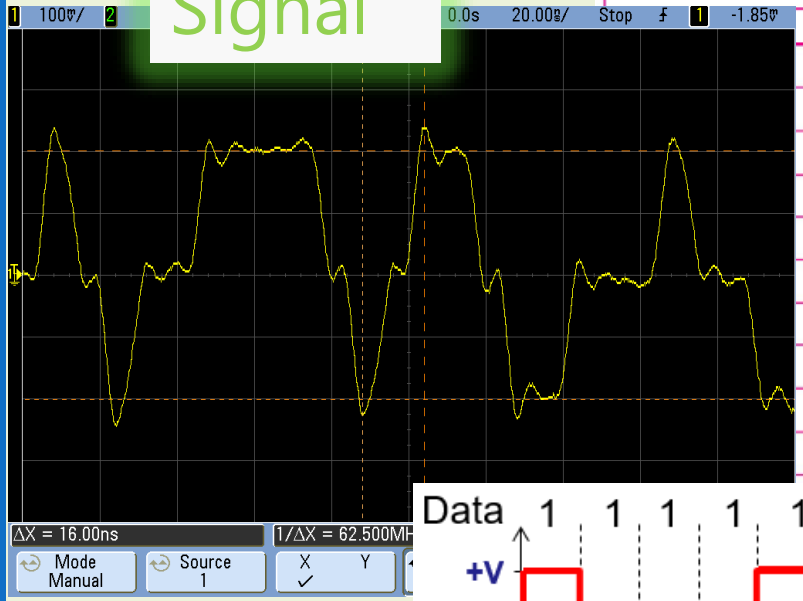
10BASE-T Ethernet Frame Analysis

Experiment 3

100BASE-TX Ethernet Signal Analysis

Ethernet
Link Integrity

One IDLE Signal

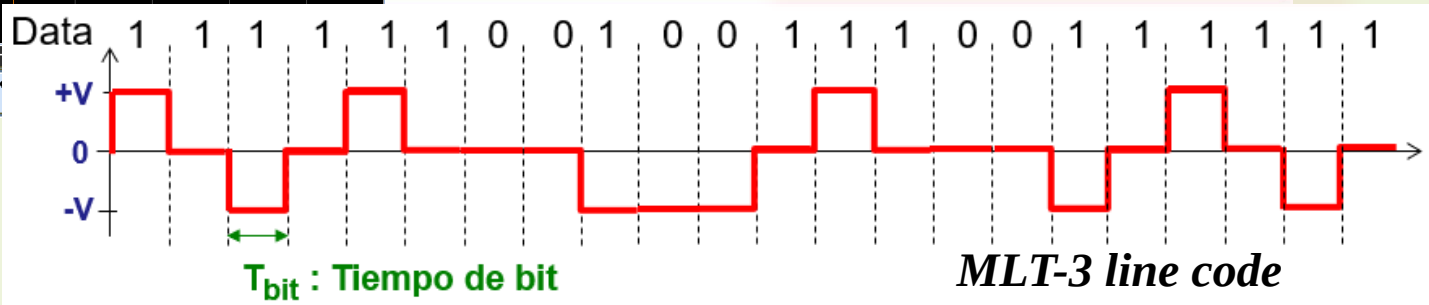


4B/5B mapping codes

Data Sequence	Encoded Sequence	Control Sequence	Encoded Sequence
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110		
0111	01111		
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

IDLE control symbol is continuously sent

Bits are scrambled

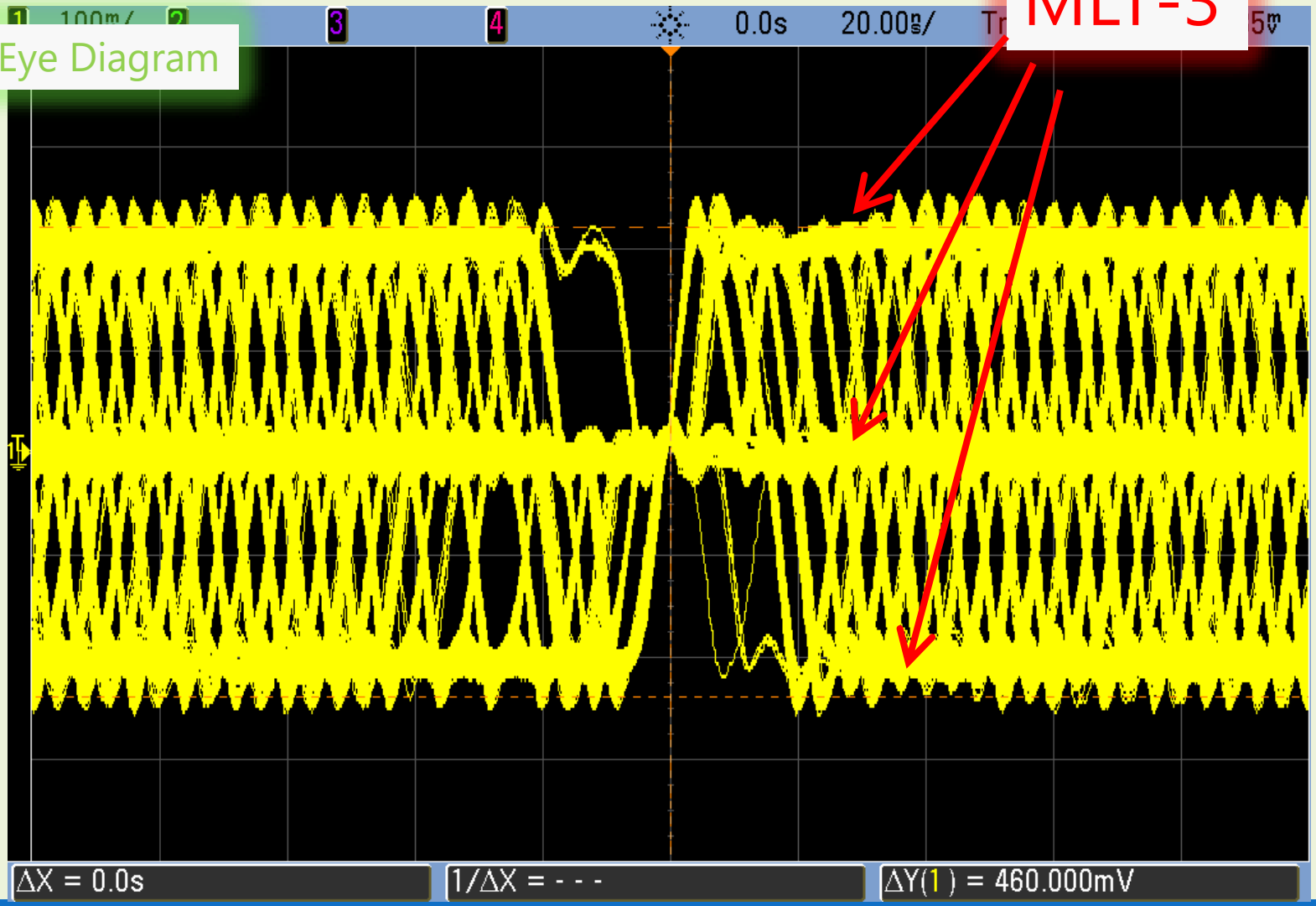


Experiment 3

100BASE-TX Ethernet Signal Analysis

Ethernet
Link Integrity

Eye Diagram



Now, send frames



Ping command from PC-B to PC-A

```
iotprofe1@huffman4:~$
iotprofe1@huffman4:~$ ping 192.168.201.103
iotprofe1@huffman4:~$
iotprofe1@huffman4:~
iotprofe1@huffman4:~
```

@IP

.1yy

yy is according to PC-A @IP

Experiment 3

100BASE-TX Ethernet Signal Analysis

Capture the

Ethernet Signal

Identify the MLT-3 signal

Measure +V, 0 and –V voltage levels

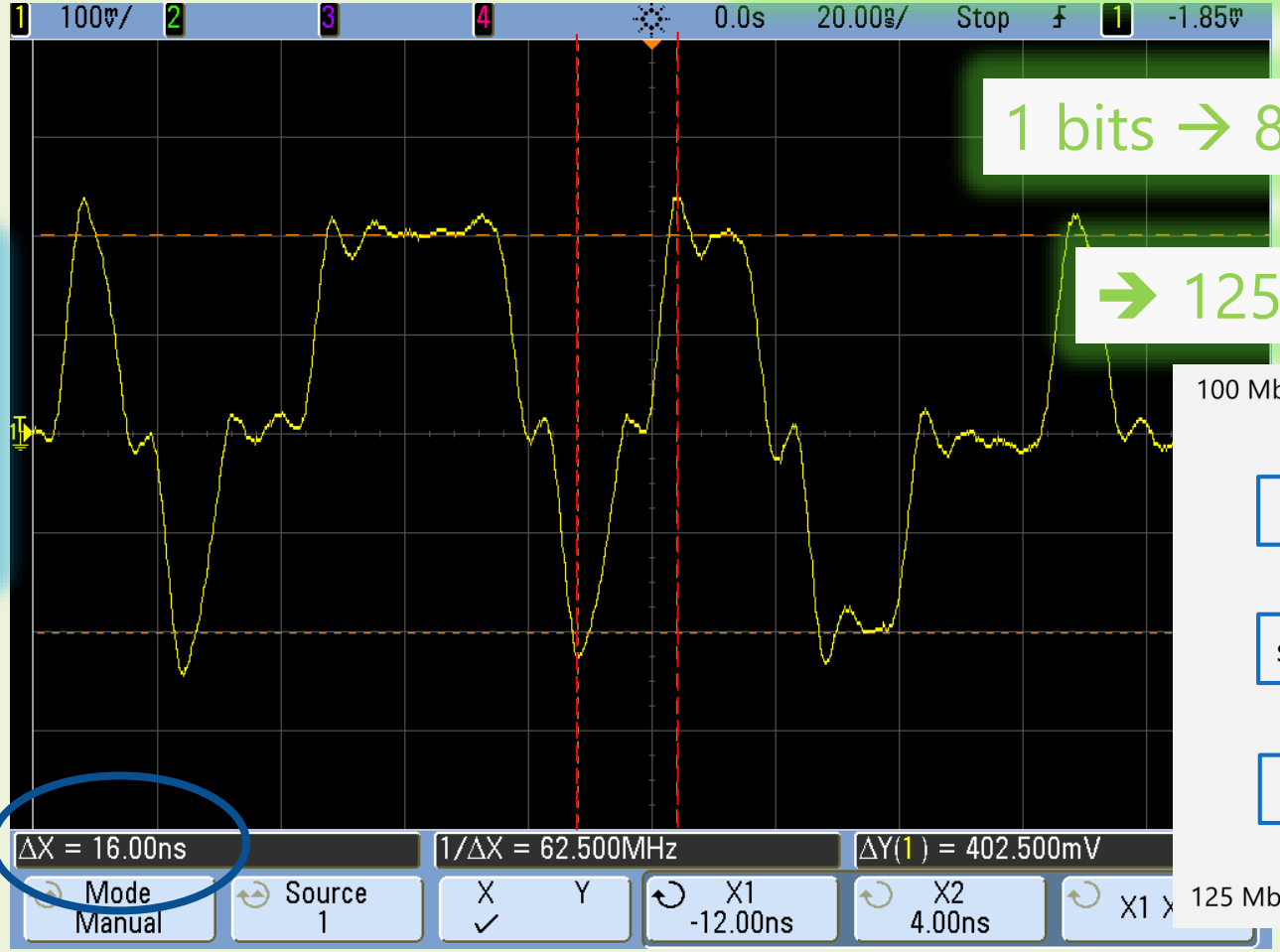
Measure the bit time and bit rate

Relate bit rate with the configured rate (100Mb/s)

Experiment 3

100BASE-TX Ethernet Signal Analysis

Ethernet bit time and rate

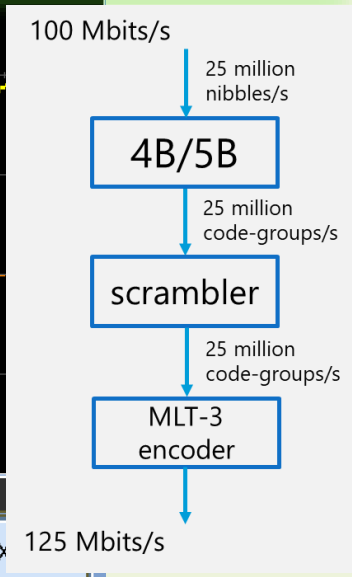


1 bits → 8 ns

→ 125 Mb/s?

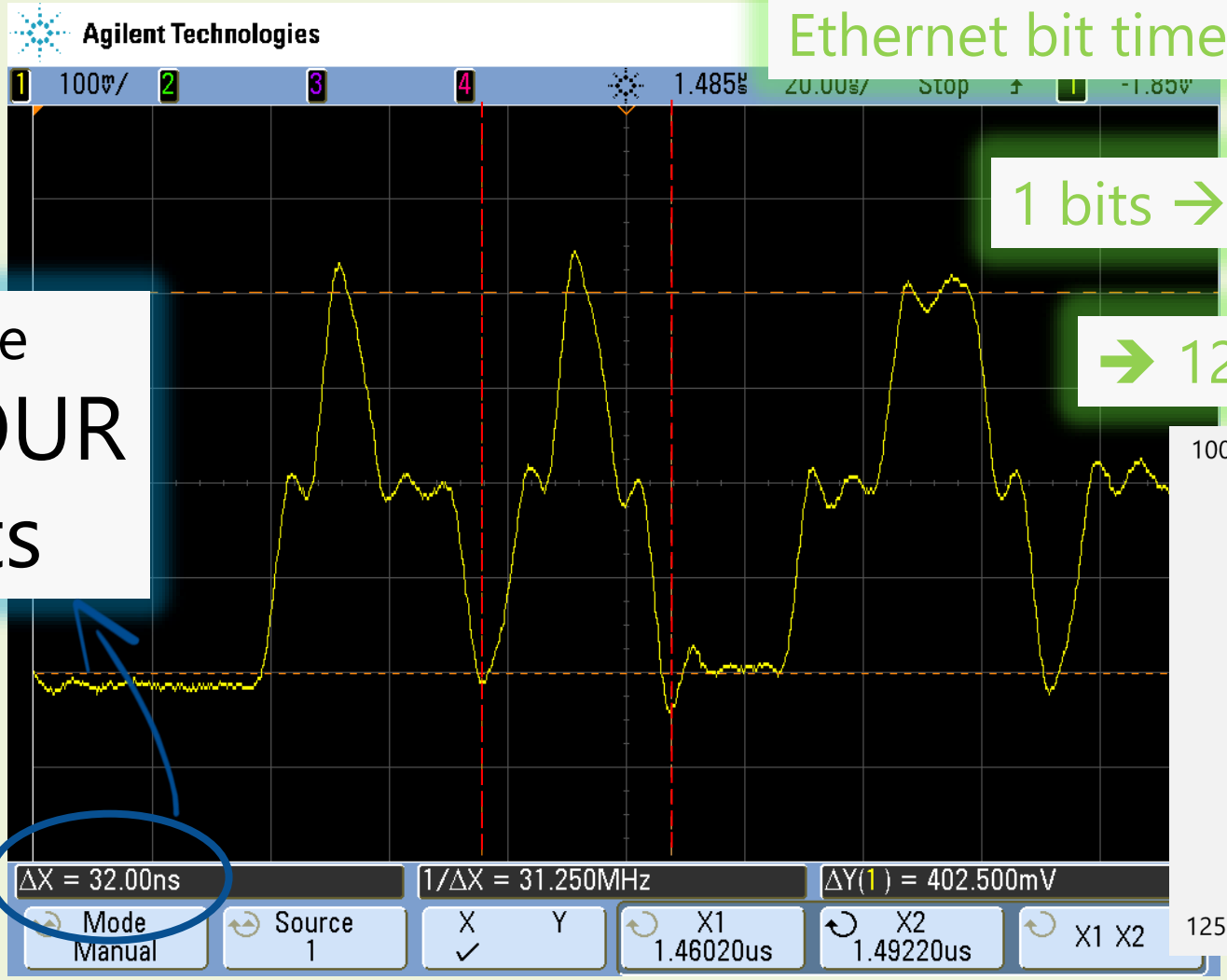
Time

TWO bits



Experiment 3

100BASE-TX Ethernet Signal Analysis



Ethernet bit time and rate

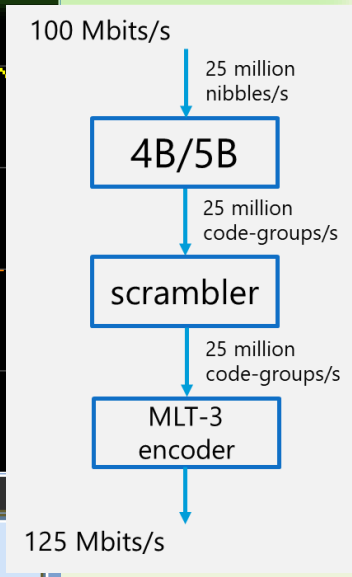
1 bits → 8 ns

→ 125 Mb/s?

Time

FOUR

bits



Experiment 3

100BASE-TX Ethernet Signal Analysis

E04

Ethernet bit time and rate



Show the measure of the bit time ✓

End of Session

Lab Rules



Lab Rules

Workstations

Leave all in a tidy and clean condition



Leave everything completely off

